# Recombinant DNA Technology

Career Avenues

## Contents

RESTRICTION AND MODIFICATION ENZYMES ..... 4
VECTORS ..... 14
cDNA AND GENOMIC DNA LIBRARY ..... 26
PROBES ..... 29
TRANSPOSONS AND GENE TARGETING ..... 31
NUCLEIC ACID LABELING ..... 14
LIBRARY SCREENING AND RFLP ..... 37
DNA SEQUENCING ..... 43
POLYMERASE CHAIN REACTION ..... 49
ELECTROPHORESIS ..... 53
DNA FINGERPRINTING ..... 55
SOUTHERN AND NORTHERN BLOTTING ..... 57
IN-SITU HYBRIDIZATION ..... 20
FROM GENOMES TO PROTEOMES ..... 21
SITE-DIRECTED MUTAGENESIS ..... 68
GENE TRANSFER TECHNOLOGIES ..... 70
GENE THERAPY ..... 73

Career Avenues

# RESTRICTION AND MODIFICATION ENZYMES

## WHAT IS GENE CLONING?

What exactly is gene cloning? The easiest way to answer this question is to follow through the steps in a gene cloning experiment:

- A fragment of DNA, containing the gene to be cloned, is inserted into a circular DNA molecule called a **vector**, to produce a **recombinant DNA molecule**.
- The vector transports the gene into a host cell, which is usually a bacterium, although other types of living cell can be used.
- Within the host cell the vector multiplies, producing numerous identical copies, not only of itself but also of the gene that it carries.
- When the host cell divides, copies of the recombinant DNA molecule are passed to the progeny and further vector replication takes place.
- After a large number of cell divisions, a colony, or **clone**, of identical host cells is produced. Each cell in the clone contains one or more copies of the recombinant DNA molecule; the gene carried by the recombinant molecule is now said to be cloned.

## RM SYSTEM

Restriction systems allow bacteria to monitor the origin of incoming DNA and to destroy it if it is recognized as foreign. Restriction endonucleases recognize specific sequences in the incoming DNA and cleave the DNA into fragments, either at specific sites or more randomly. When the incoming DNA is a bacteriophage genome, the effect is to reduce the efficiency of plating, i.e. to reduce the number of plaques formed in plating tests. The phenomena of restriction and modification were well illustrated and studied by the behaviour of phage λ on two *E. coli* host strains.

Restriction-Modification (RM) system occurs in many bacterial species and constitute a defense mechanism against the introduction of foreign DNA into the cell. They consist of two components:

- Restriction endonuclease: Which recognizes a short, symmetrical DNA sequence and hydrolyse DNA backbone in each strand at a specific sites within that sequence - so that the foreign DNA will hence be degraded to relatively short fragments.

Career Avenues

- Methylase: Adds a methyl group to a C or A base within the same recognition sequences in the cellular DNA - This modification renders the host DNA resistant to degradation by the endonuclease.

## TYPES OF RESTRICTION AND MODIFICATION (R-M) SYSTEM

There are three major classes of restriction endonucleases. Their grouping is based on the types of sequences recognized, the nature of the cut made in the DNA, and the enzyme structure. Type I and III restriction endonucleases are not useful for gene cloning because they cleave DNA at sites other than the recognition sites and thus cause random cleavage patterns. In contrast, type II endonucleases are widely used for mapping and reconstructing DNA *in vitro* because they recognize specific sites and cleave just at these sites (Table).

In addition, the type II endonuclease and methylase activities are usually separate, single subunit enzymes. Although the two enzymes recognize the same target sequence, they can be purified separately from each other.

| Class | Abundance | Recognition site | Composition | Use in recombinant DNA research |
|---|---|---|---|---|
| Type I | Less common than type II | Cut both strands at a nonspecific location > 1000 bp away from recognition site | Three-subunit complex: individual recognition, endonuclease, and methylase activities | Not useful |
| Type II | Most common | Cut both strands at a specific, usually palindromic, recognition site (4–8 bp) | Endonuclease and methylase are separate, single-subunit enzymes | Very useful |
| Type III | Rare | Cleavage of one strand only, 24–26 bp downstream of the 3′ recognition site | Endonuclease and methylase are separate two-subunit complexes with one subunit in common | Not useful |

The type I systems were the first to be characterized and a typical example is that from *E. coli* K12. The active enzyme consists of two restriction subunits, two modification (methylation) subunits and one recognition subunit. These subunits are the products of the *hsd*R, *hsd*M and *hsd*S genes. The methylation and cutting reactions both require ATP and *S*-adenosylmethionine as cofactors. The recognition sequences are quite long with no recognizable features such as symmetry. The enzyme also cuts unmodified DNA at some distance from the recognition sequence. However, because the Methylation reaction is performed by the same enzyme which mediates cleavage, the target DNA may be modified before it is cut. These features mean that type I systems are of little value for gene manipulation. However, their presence in *E. coli* strains can affect recovery of recombinants.

Type III enzymes have symmetrical recognition sequences but otherwise resemble type I systems and are of little value. Most of the useful R-M systems are of type II. They have a number of advantages over type I and III systems. First, restriction and modification are mediated by separate enzymes so it is possible to cleave DNA in the absence of modification. Secondly, the restriction activities do not require cofactors such as ATP or *S*-adenosylmethionine, making them

easier to use. Most important of all, type II enzymes recognize a defined, usually symmetrical, sequence *and cut within it*. Many of them also make a staggered break in the DNA and the usefulness of this will become apparent. Although type IIs systems have similar cofactors and macromolecular structure to those of type II systems, the fact that restriction occurs at a distance from the recognition site limits their usefulness.

## TYPE II RESTRICTION ENDONUCLEASES

The central feature of type II restriction endonucleases is that each enzyme has a specific recognition sequence at which it cuts a DNA molecule. A particular enzyme cleaves DNA at the recognition sequence and nowhere else. For example, the restriction endonuclease called *Pvu*I (isolated from *Proteus vulgaris*) cuts DNA only at the hexanucleotide CGATCG.

In contrast, a second enzyme from the same bacterium, called *Pvu*II, cuts at a different hexanucleotide, in this case CAGCTG.

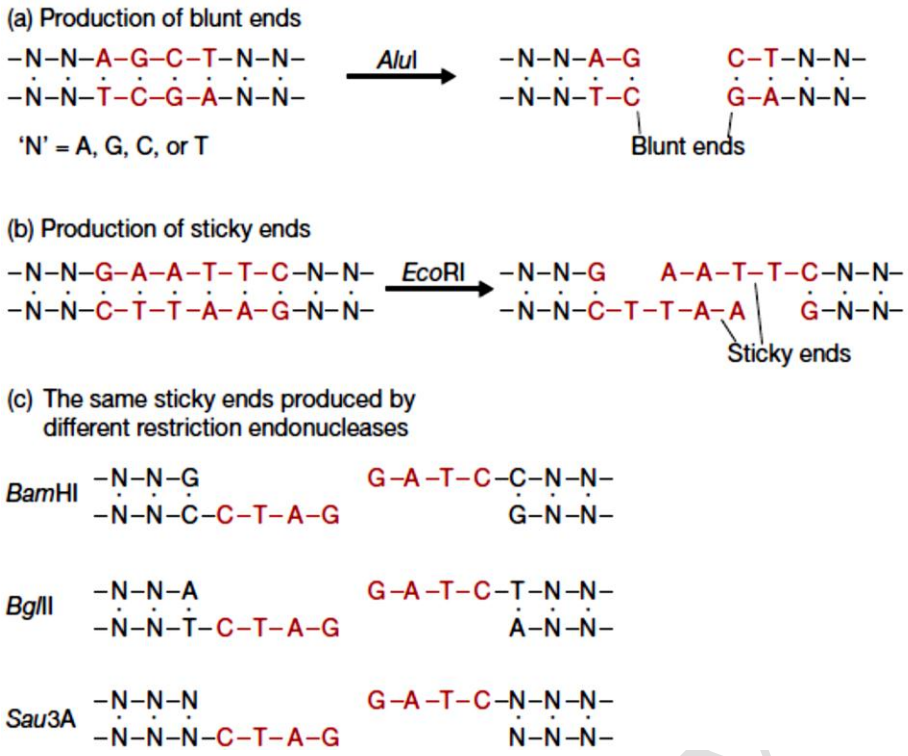
Many restriction endonucleases recognize hexanucleotide target sites, but others cut at four, five, eight, or even longer nucleotide sequences. *Sau*3A (from *Staphylococcus aureus* strain 3A) recognizes GATC, and *Alu*I (*Arthrobacter luteus*) cuts at AGCT. There are also examples of restriction endonucleases with degenerate recognition sequences, meaning that they cut DNA at any one of a family of related sites. *Hin*fI (*Haemophilus influenzae* strain Rf), for instance, recognizes GANTC, so cuts at GAATC, GATTC, GAGTC, and GACTC.

## BLUNT ENDS AND STICKY ENDS

The exact nature of the cut produced by a restriction endonuclease is of considerable importance in the design of a gene cloning experiment. Many restriction endonucleases make a simple double-stranded cut in the middle of the recognition sequence, resulting in a **blunt end** or **flush end**. *Pvu*II and *Alu*I are examples of blunt end cutters. Other restriction endonucleases cut DNA in a slightly different way. With these enzymes the two DNA strands are not cut at exactly the same position. Instead the cleavage is staggered, usually by two or four nucleotides, so that the resulting DNA fragments have short single-stranded overhangs at each end. These are called sticky or cohesive ends, as base pairing between them can stick the DNA molecule back together.

One important feature of sticky end enzymes is that restriction endonucleases with different recognition sequences may produce the same sticky ends. *Bam*HI (recognition sequence GGATCC) and *Bgl*II (AGATCT) are examples—both produce GATC sticky ends (Figure 4.9c). The same sticky end is also produced by *Sau*3A, which recognizes only the tetranucleotide GATC. Fragments of DNA produced by cleavage with either of these enzymes can be joined to each other, as each fragment carries a complementary sticky end.

Career Avenues




## THE FREQUENCY OF RECOGNITION SEQUENCES IN A DNA MOLECULE:

The number of recognition sequences for a particular restriction endonuclease in a DNA molecule of known length can be calculated mathematically. A tetranucleotide sequence (e.g., GATC) should occur once every $4^4$ =256 nucleotides, and a hexanucleotide (e.g., GGATCC) once every $4^6$ =4096 nucleotides. These calculations assume that the nucleotides are ordered in a random fashion and that the four different nucleotides are present in equal proportions (i.e., the GC content =50%). In practice, neither of these assumptions is entirely valid. For example, the e DNA molecule, at 49 kb, should contain about 12 sites for a restriction endonuclease with a hexanucleotide recognition sequence. In fact, many of these recognition sites occur less frequently (e.g., six for *Bgl*II, five for *Bam*HI, and only two for *Sal*I), a reflection of the fact that the GC content for e is rather less than 50%.

## THE RANGE OF DNA MANIPULATIVE ENZYMES:

DNA manipulative enzymes can be grouped into four broad classes, depending on the type of reaction that they catalyze:

- **Nucleases** are enzymes that cut, shorten, or degrade nucleic acid molecules.
- **Ligases** join nucleic acid molecules together.
- **Polymerases** make copies of molecules.
- **Modifying enzymes** remove or add chemical groups.

Career Avenues

## NUCLEASES

Nucleases degrade DNA molecules by breaking the phosphodiester bonds that link one nucleotide to the next in a DNA strand. There are two different kinds of nuclease:

- **Exonucleases** remove nucleotides one at a time from the end of a DNA molecule.
- **Endonucleases** are able to break internal phosphodiester bonds within a DNA molecule.

The main distinction between different exonucleases lies in the number of strands that are degraded when a double-stranded molecule is attacked. The enzyme called Bal31 is an example of an exonuclease that removes nucleotides from both strands of a double-stranded molecule. The greater the length of time that Bal31 is allowed to act on a group of DNA molecules, the shorter the resulting DNA fragments will be. In contrast, enzymes such as *E. coli* exonuclease III degrade just one strand of a double-stranded molecule, leaving single-stranded DNA as the product.

The same criterion can be used to classify endonucleases. S1 endonuclease (from the fungus *Aspergillus oryzae*) only cleaves single strands, whereas deoxyribonuclease I (DNase I), which is prepared from cow pancreas, cuts both single and double-stranded molecules. DNase I is non-specific in that it attacks DNA at any internal phosphodiester bond, so the end result of prolonged DNase I action is a mixture of mononucleotides and very short oligonucleotides.

## DNA LIGASES

Ligases catalyze the formation of phosphodiester bonds between the directly adjacent 3'-hydroxyl and 5' phosphoryl termini of nucleic acid molecules. The substrates may be DNA or RNA, and the cofactors that generate high-energy intermediates in the reaction may be ATP or $NAD^+$, depending on the type of ligase.

*E. coli* and phage T4 encode an enzyme, DNA ligase, which seals single-stranded nicks between adjacent nucleotides in a duplex DNA chain. Although the reactions catalysed by the enzymes of *E. coli* and T4-infected *E. coli* are very similar, they differ in their cofactor requirements. The T4 enzyme requires ATP, while the *E. coli* enzyme requires $NAD^+$. In each case the cofactor is split and forms an enzyme– AMP complex. The complex binds to the nick, which must expose a 5′-phosphate and 3′-OH group, and makes a covalent bond in the phosphodiester chain.

The optimum temperature for ligation of nicked DNA is 37°C, but at this temperature the hydrogen bonded join between the sticky ends is unstable. *Eco*RI-generated termini associate through only four AT base pairs and these are not sufficient to resist thermal disruption at such a high temperature. The optimum temperature for ligating the cohesive termini is therefore a compromise between the rate of enzyme action and association of the termini, and has been found experimentally to be in the range 4–15°C.

The *E. coli* DNA ligase will not catalyse blunt ligation except under special reaction conditions of macromolecular crowding.

Career Avenues

## POLYMERASES

DNA polymerases are enzymes that synthesize a new strand of DNA complementary to an existing DNA or RNA template (Figure). Most polymerases can function only if the **template** possesses a double-stranded region that acts as a **primer** for initiation of polymerization.

Four types of DNA polymerase are used routinely in genetic engineering. The first is DNA polymerase I, which is usually prepared from *E. coli*. This enzyme attaches to a short single-stranded region (or **nick**) in a mainly double-stranded DNA molecule, and then synthesizes a completely new strand, degrading the existing strand as it proceeds. DNA polymerase I is therefore an example of an enzyme with a dual activity—DNA polymerization and DNA degradation.

The polymerase and nuclease activities of DNA polymerase I are controlled by different parts of the enzyme molecule. The nuclease activity is contained in the first 323 amino acids of the polypeptide, so removal of this segment leaves a modified enzyme that retains the polymerase function but is unable to degrade DNA. This modified enzyme, called the **Klenow fragment**, can still synthesize a complementary DNA strand on a single-stranded template, but as it has no nuclease activity it cannot continue the synthesis once the nick is filled in.

Several other enzymes—natural polymerases and modified versions—have similar properties to the Klenow fragment. The major application of these polymerases is in DNA sequencing. The *Taq* DNA polymerase used in the polymerase chain reaction (PCR) is the DNA polymerase I enzyme of the bacterium *Thermus aquaticus*. This organism lives in hot springs, and many of its enzymes, including the *Taq* DNA polymerase, are thermostable, meaning that they are resistant to denaturation by heat treatment. This is the special feature of *Taq* DNA polymerase that makes it suitable for PCR, because if it was not thermostable it would be inactivated when the temperature of the reaction is raised to 94°C to denature the DNA.

The final type of DNA polymerase that is important in genetic engineering is **reverse transcriptase**, an enzyme involved in the replication of several kinds of virus. Reverse transcriptase is unique in that it uses as a template not DNA but RNA. The ability of this enzyme to synthesize a DNA strand complementary to an RNA template is central to the technique called complementary DNA (cDNA) cloning.

Career Avenues

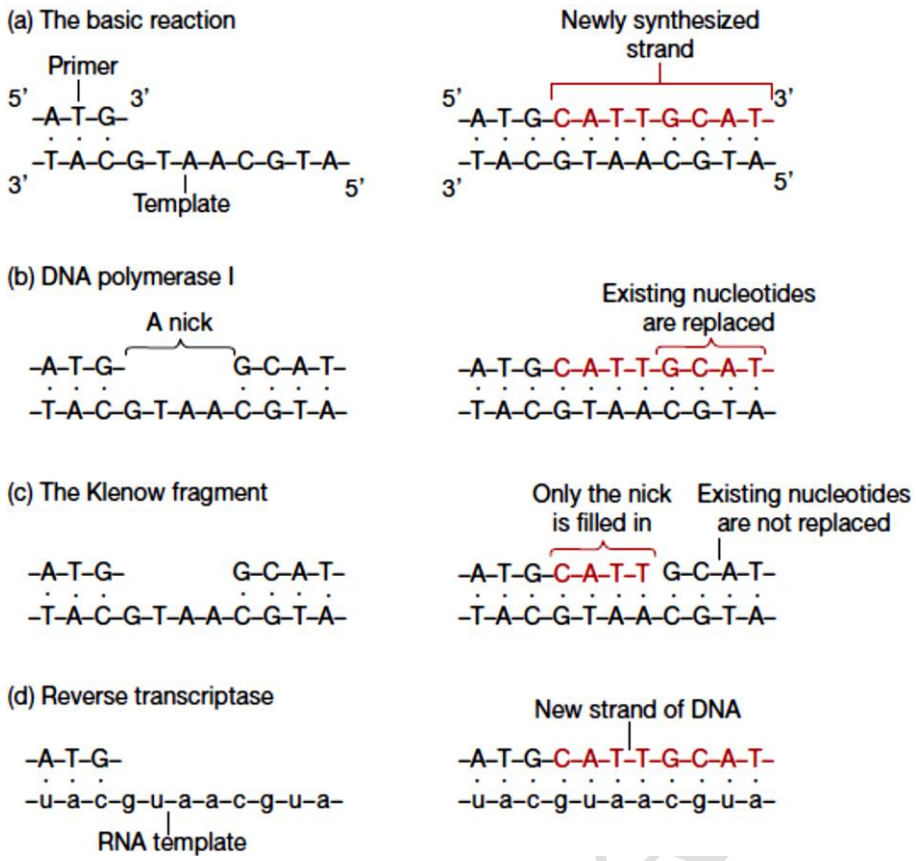



**Figure:** The reactions catalyzed by DNA polymerases. (a) The basic reaction: a new DNA strand is synthesized in the 5′- to 3′- direction. (b) DNA polymerase I, which initially fills in nicks but then continues to synthesize a new strand, degrading the existing one as it proceeds. (c) The Klenow fragment, which only fills in nicks. (d) Reverse transcriptase, which uses a template of RNA.

## DNA MODIFYING ENZYMES

There are numerous enzymes that modify DNA molecules by addition or removal of specific chemical groups. The most important are as follows:

- **Alkaline phosphatase** (from *E. coli*, calf intestinal tissue, or arctic shrimp), which removes the phosphate group present at the 5′- terminus of a DNA molecule.
- **Polynucleotide kinase** (from *E. coli* infected with T4 phage), which has the reverse effect to alkaline phosphatase, adding phosphate groups onto free 5′ termini.
- **Terminal deoxynucleotidyl transferase** (from calf thymus tissue), which adds one or more deoxyribonucleotides onto the 3′ terminus of a DNA molecule.

Career Avenues

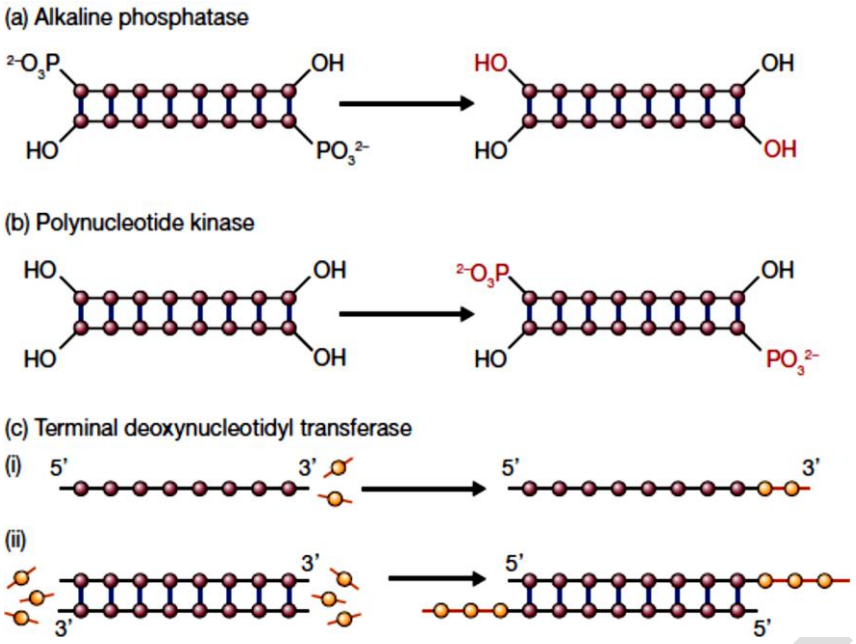
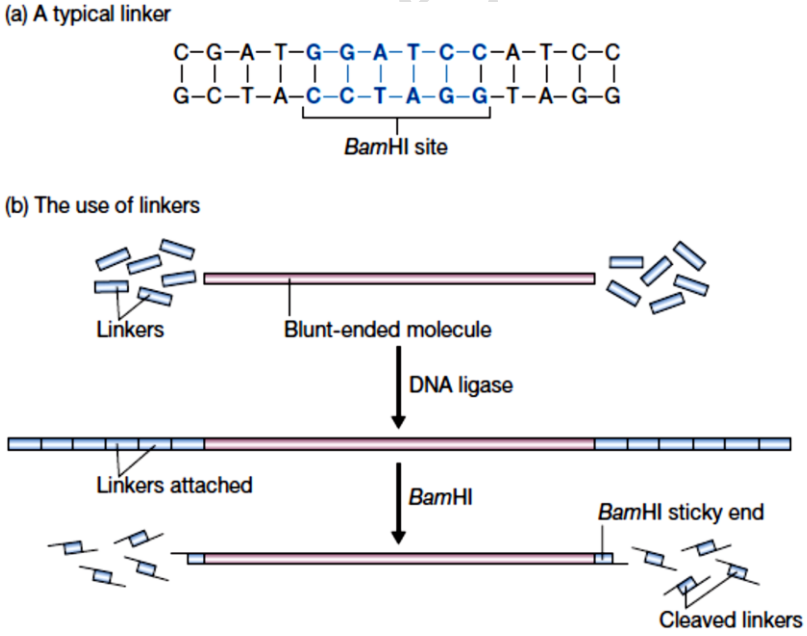



**Figure:** The reactions catalyzed by DNA modifying enzymes. (a) Alkaline phosphatase, which removes 5'-phosphate groups. (b) Polynucleotide kinase, which attaches 5'-phosphate groups. (c) Terminal deoxynucleotidyl transferase, which attaches deoxyribonucleotides to the 3' termini of polynucleotides in either (i) single-stranded or (ii) double-stranded molecules.

## SOME IMPORTANT NOTES

## LINKERS

The first of these methods involves the use of **linkers**. These are short pieces of double stranded DNA, of known nucleotide sequence, that are synthesized in the test tube.




Career Avenues

**Figure:** Linkers and their use: (a) the structure of a typical linker; (b) the attachment of linkers to a blunt-ended molecule.

A typical linker is shown in Figure. It is blunt-ended, but contains a restriction site, *Bam*HI in the example shown. DNA ligase can attach linkers to the ends of larger blunt ended DNA molecules. Although a blunt end ligation, this particular reaction can be performed very efficiently because synthetic oligonucleotides, such as linkers, can be made in very large amounts and added into the ligation mixture at a high concentration. More than one linker will attach to each end of the DNA molecule, producing the chain structure. However, digestion with *Bam*HI cleaves the chains at the recognition sequences, producing a large number of cleaved linkers and the original DNA fragment, now carrying *Bam*HI sticky ends. This modified fragment is ready for ligation into a cloning vector restricted with *Bam*HI.

## ADAPTORS

**Adaptors**, like linkers, are short synthetic oligonucleotides. But unlike linkers, an adaptor is synthesized so that it already has one sticky end. The idea is of course to ligate the blunt end of the adaptor to the blunt ends of the DNA fragment, to produce a new molecule with sticky ends. This may appear to be a simple method but in practice a new problem arises. The sticky ends of individual adaptor molecules could base pair with each other to form dimers, so that the new DNA molecule is still blunt-ended. The sticky ends could be recreated by digestion with a restriction endonuclease, but that would defeat the purpose of using adaptors in the first place.

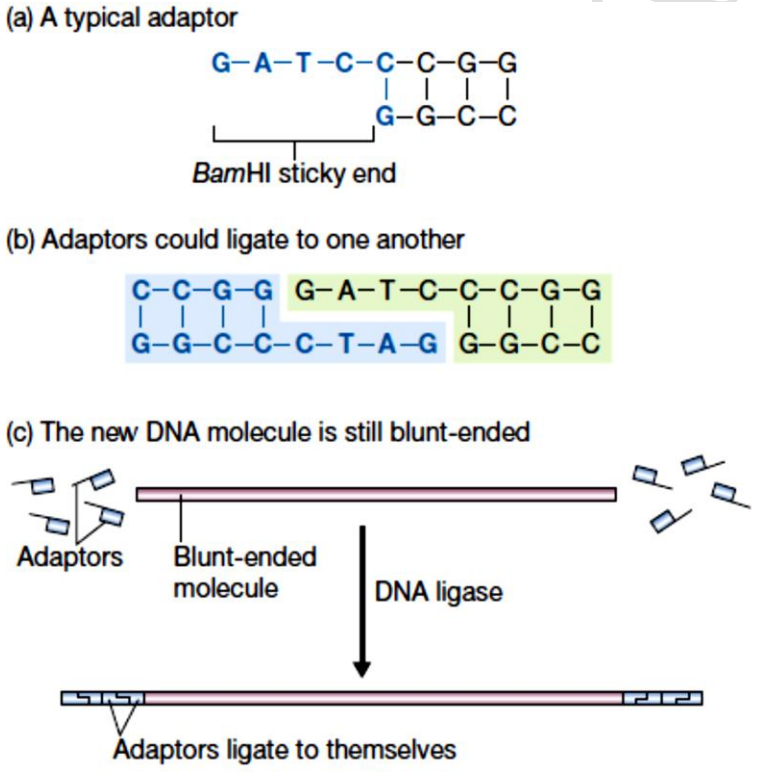


**Figure:** Adaptors and the potential problem with their use. (a) A typical adaptor. (b) Two adaptors could ligate to one another to produce a molecule similar to a linker, so that (c) after ligation of adaptors a blunt-ended molecule is still blunt-ended and the restriction step is still needed.

## HOMOPOLYMER TAILING

A general method for joining DNA molecules makes use of the annealing of complementary homopolymer sequences. Thus, by adding oligo(dA) sequences to the 3'- ends of one population of DNA molecules and oligo(dT) blocks to the 3'- ends of another population, the two types of molecule can anneal to form mixed dimeric circles. An enzyme purified from calf thymus, terminal deoxynucleotidyltransferase, provides the means by which the homopolymeric extensions can be synthesized, for if presented with a single deoxynucleotide triphosphate it will repeatedly add nucleotides to the 3'OH termini of a population of DNA molecules. DNA with exposed 3'OH groups, such as arise from pretreatment with phage λ exonuclease or restriction with an enzyme such as *Pst*I, is a very good substrate for the transferase. However, conditions have been found in which the enzyme will extend even the shielded 3' -OH of 5' cohesive termini generated by *Eco*RI.

Career Avenues

# VECTORS

The basic procedure of molecular cloning involves a series of steps. First, the DNA fragments to be cloned are generated by using restriction endonucleases. Second, the fragments produced by digestion with restriction enzymes are ligated to other DNA molecules that serve as vectors.

Vectors can replicate autonomously (independent of host genome replication) in host cells and facilitate the manipulation of the newly created recombinant DNA molecule. Third, the recombinant DNA molecule is transferred to a host cell. Within this cell, the recombinant DNA molecule replicates, producing dozens of identical copies known as clones. As the host cells replicate, the recombinant DNA is passed on to all progeny cells, creating a population of identical cells, all carrying the cloned sequence. Finally, the cloned DNA segments can be recovered from the host cell, purified, and analyzed in various ways.

## VECTOR DNA

Cloning vectors are carrier DNA molecules. Four important features of all cloning vectors are that they:

- Can independently replicate themselves and the foreign DNA segments they carry;
- Contain a number of unique restriction endonuclease cleavage sites that are present only once in the vector;
- Carry a selectable marker (usually in the form of antibiotic resistance genes or genes for enzymes missing in the host cell) to distinguish host cells that carry vectors from host cells that do not contain a vector; and
- Are relatively easy to recover from the host cell. There are many possible choices of vector depending on the purpose of cloning. The greatest variety of cloning vectors has been developed for use in the bacterial host *E. coli*.

Thus, the first practical skill generally required by a molecular biologist is the ability to grow pure cultures of bacteria.

## CHOICE OF VECTOR IS DEPENDENT ON INSERT SIZE AND APPLICATION

The classic cloning vectors are plasmids, phages, and cosmids, which are limited to the size insert they can accommodate, taking up to 10, 20, and 45 kb, respectively. The feature of plasmids and phages and their use as cloning vectors will be discussed in more detail in later sections. A cosmid is a plasmid carrying a phage λ *cos* site, allowing it to be packaged into a phage head. Cosmids infect a host bacterium as do phages, but replicate like plasmids and the host cells are not lysed. Mammalian genes are often greater than 100 kb in size, so originally

Career Avenues

there were limitations in cloning complete gene sequences. Vectors engineered more recently have circumvented this problem by mimicking the properties of host cell chromosomes. This new generation of artificial chromosome vectors includes bacterial artificial chromosomes (BACs), yeast artificial chromosomes (YACs), and mammalian artificial chromosomes (MACs).

## PRINCIPAL FEATURES AND APPLICATIONS OF DIFFERENT CLONING VECTOR SYSTEMS

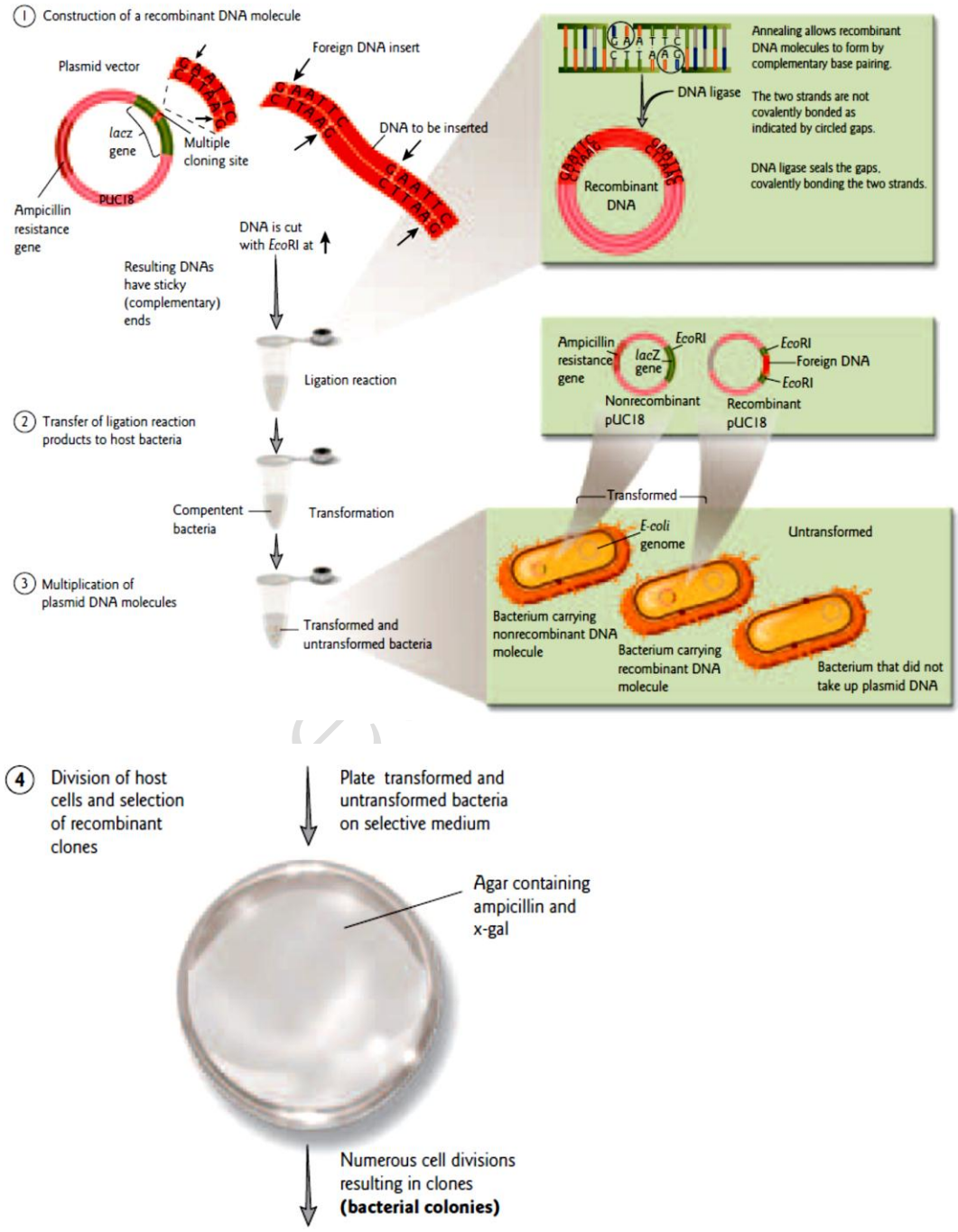
| Vector | Basis | Size limits of insert | Major application |
|---|---|---|---|
| **Plasmid** | Naturally occuring multicopy plasmids | ≤ 10 kb | Subcloning and downstream manipulation, cDNA cloning and expression assays |
| **Phage** | Bacteriophage λ | 5–20 kb | Genomic DNA cloning, cDNA cloning, and expression libraries |
| **Cosmid** | Plasmid containing a bacteriophage λ *cos* site | 35–45 kb | Genomic library construction |
| **BAC (bacterial artificial chromosome)** | *Escherichia coli* F factor plasmid | 75–300 kb | Analysis of large genomes |
| **YAC (yeast artificial chromosome)** | *Saccharomyces cerevisiae* centromere, telomere, and autonomously replicating sequence | 100–1000 kb (1 Mb) | Analysis of large genomes, YAC transgenic mice |
| **MAC (mammalian artificial chromosome)** | Mammalian centromere, telomere, and origin of replication | 100 kb to > 1 Mb | Under development for use in animal biotechnology and human gene therapy |

## PLASMID DNA AS A VECTOR

Plasmids are naturally occurring extrachromosomal double-stranded circular DNA molecules that carry an origin of replication and replicate autonomously within bacterial cells. The plasmid vector pBR322, constructed in 1974, was one of the first genetically engineered plasmids to be used in recombinant DNA. Plasmids are named with a system of uppercase letters and numbers, where the lowercase "p" stands for "plasmid." In the case of pBR322, the BR identifies the original constructors of the vector (Bolivar and Rodriquez), and 322 is the identification number of the specific plasmid. These early vectors were often of low copy number, meaning that they replicate to yield only one or two copies in each cell.

pUC18, the vector is a derivative of pBR322. This is a "high copy number" plasmid (~500 copies per bacterial cell). Plasmid vectors are modified to contain a specific antibiotic resistance gene and a multiple cloning site (also called the polylinker region) which has a number of unique target sites for restriction endonucleases. Cutting the circular plasmid vector with one of these enzymes results in a single cut, creating a linear plasmid. A foreign DNA molecule, referred to as the "insert," cut with the same enzyme, can then be joined to the vector in a ligation reaction. Ligations of the insert to vector are not 100% productive, because the two ends of a plasmid vector can be readily ligated together, which is called self-ligation. The degree of self-ligation

can be reduced by treatment of the vector with the enzyme phosphatase, which removes the terminal 5′-phosphate. When the 5′-phosphate is removed from the plasmid it cannot be recircularized by ligase, since there is nothing with which to make a phosphodiester bond. But, if the vector is joined with a foreign insert, the 5′-phosphate is provided by the foreign DNA. Another strategy involves using two different restriction endonuclease cutting sites with non-complementary sticky ends. This inhibits self-ligation and promotes annealing of the foreign DNA in the desired orientation within the vector.



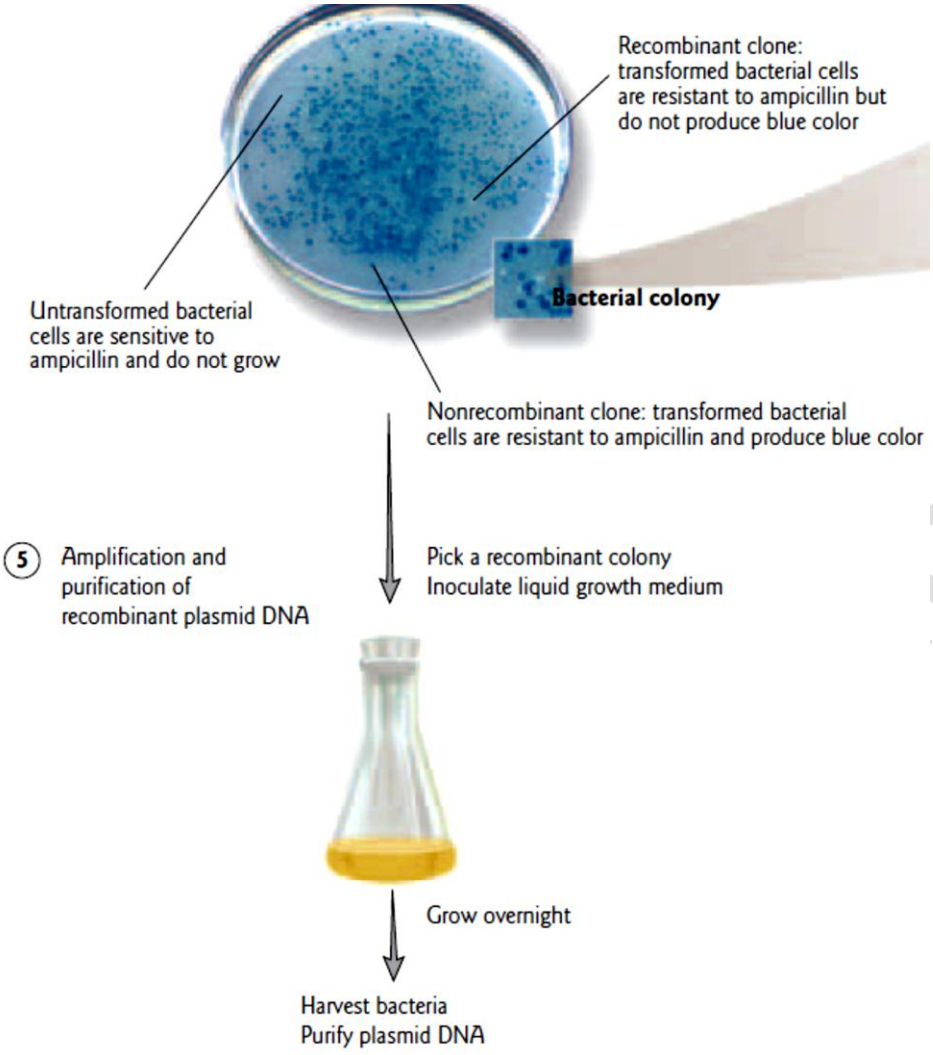



**Figure: Molecular cloning using a plasmid vector.** Molecular cloning using a plasmid vector involves five major steps. (1) Construction of a recombinant DNA molecule. In this example, vector DNA (the plasmid pUC18) and the foreign DNA insert are cleaved with *Eco*RI and mixed together in a ligation reaction containing DNA ligase. pUC18 carries the ampicillin resistance gene and has a large number of restriction sites comprising a multiple cloning site within a selectable marker gene. (2) Transfer of ligation reaction products to host bacteria. Competent *E. coli* are transformed with ligation reaction products. Any DNA that remains linear will not be taken up by the host bacteria. (3) Multiplication of plasmid DNA molecules. Within each transformed host bacterium, there is autonomous multiplication of plasmid DNA. Each bacterium may contain as many as 500 copies of pUC18. Some bacteria in the mixture will be untransformed (not carrying either recombinant or nonrecombinant plasmid DNA). (4) Division of host cells and selection of recombinant clones by blue-white screening. Bacteria are plated on a selective agar medium containing the antibiotic ampicillin and X-gal. If foreign DNA is inserted into the multiple cloning site, then the *lacZ'* coding region is disrupted and the N-terminal portion of β-galactosidase is not produced. Since there is no functional β-galactosidase in the bacteria, the substrate X-gal remains colorless, and the bacterial colony containing recombinant plasmid DNA appears white, thus allowing

the direct identification of colonies carrying cloned DNA inserts. If there is no insertion of foreign DNA in the multiple cloning site, then the *lac*Z′ gene is intact and enzymatically active β-galactosidase is produced. The bacterial colonies containing nonrecombinant plasmid DNA thus appear blue. (5) Amplification and purification of recombinant plasmid DNA. A recombinant colony is used to inoculate liquid growth medium. After growing the bacteria overnight, the culture is harvested, bacterial cells are lysed, and the plasmid DNA is purified away from other cellular components.

## TRANSFORMATION: TRANSFER OF RECOMBINANT PLASMID DNA TO A BACTERIAL HOST

The ligation reaction mixture of recombinant and nonrecombinant DNA described in the preceding section is introduced into bacterial cells in a process called transformation. The traditional method is to incubate the cells in a concentrated calcium salt solution to make their membranes leaky. The permeable "competent" cells are then mixed with DNA to allow entry of the DNA into the bacterial cell. Alternatively, a process called electroporation can be used that drives DNA into cells by a strong electric current. Successfully transformed bacteria will carry either recombinant or nonrecombinant plasmid DNA. Multiplication of the plasmid DNA occurs within each transformed bacterium. A single bacterial cell placed on a solid surface (agar plate) containing nutrients can multiply to form a visible colony made of millions of identical cells. As the host cell divides, the plasmid vectors are passed on to progeny, where they continue to replicate. Numerous cell divisions of a single transfomed bacteria result in a clone of cells (visible as a bacterial colony) from a single parental cell. This step is where "cloning" got its name. The cloned DNA can then be isolated from the clone of bacterial cells.

## RECOMBINANT SELECTION

What needs to be included in the medium for plating cells so that nontransformed bacterial cells are not able to grow at all? The answer depends on the particular vector, but in the case of pUC18, the vector carries a selectable marker gene for resistance to the antibiotic ampicillin. Ampicillin, a derivative of penicillin, blocks synthesis of the peptidoglycan layer that lies between the inner and outer cell membranes of *E. coli*. Ampicillin does not affect existing cells with intact cell envelopes but kills dividing cells as they synthesize new peptidoglycan. The ampicillin resistance genes carried by the recombinant plasmids produce an enzyme, β-lactamase, that cleaves a specific bond in the four-membered ring (β-lactam ring) in the ampicillin molecule that is essential to its antibiotic action. If the plasmid vector is introduced into a plasmid free antibiotic-sensitive bacterial cell, the cell becomes resistant to ampicillin. Nontransformed cells contain no pUC18 DNA, therefore they will not be antibiotic-resistant, and their growth will be inhibited on agar containing ampicillin. Transformed bacterial cells may contain either nonrecombinant pUC18 DNA (self ligated vector only) or recombinant pUC18 DNA (vector containing foreign DNA insert). Both types of transformed bacterial cells will be ampicillin-resistant.

Career Avenues

**Table: Some commonly used antibiotics and antibiotic resistance genes**

| Antibiotic | Mode of action | Resistance gene |
|---|---|---|
| Ampicillin | Inhibits bacterial cell wall synthesis by disrupting peptidoglycan cross-linking | β-Lactamase (*amp*$^r$) gene product is secreted and hydrolyzes ampicillin |
| Tetracycline | Inhibits binding of aminoacyl tRNA to the 30S ribosomal subunit | *tet*$^r$ gene product is membrane bound and prevents tetracycline accumulation by an efflux mechanism |
| Kanamycin | Inactivates translation by interfering with ribosome function | Neomycin or aminoglycoside phosphotransferase (*neo*$^r$) gene product inactivates kanamycin by phosphorylation |

## BLUE-WHITE SCREENING

To distinguish nonrecombinant from recombinant transformants, blue-white screening or "*lac* selection" (also called α-complementation) can be used with this particular vector. Bacterial colonies are grown on selective medium containing ampicillin and a colorless chromogenic compound called X-gal, for short (5-bromo-4-chloro-3-indolyl-β-d-galactoside). pUC18 carries a portion of the *lacZ* gene (called *lacZ ′*) that encodes the first 146 amino acids for the enzyme β-galactosidase. The multiple cloning site resides in the coding region. If the *lacZ'* region is not interrupted by inserted DNA, the amino terminal portion of β-galactosidase is synthesized. Importantly, an *E. coli* deletion mutant strain is used (e.g. DH5α strain of *E. coli* cell) that harbors a mutant sequence of *lacZ* that encodes only the carboxyl end of β-galactosidase (*lacZ*′ ΔM15). Both the plasmid and host *lacZ* fragments encode nonfunctional proteins. However, by α-complementation the two partial proteins can associate and form a functional enzyme. When present, the enzyme β-galactosidase catalyzes hydrolysis of X-gal, converting the colorless substrate into a blue-colored product.

Career Avenues

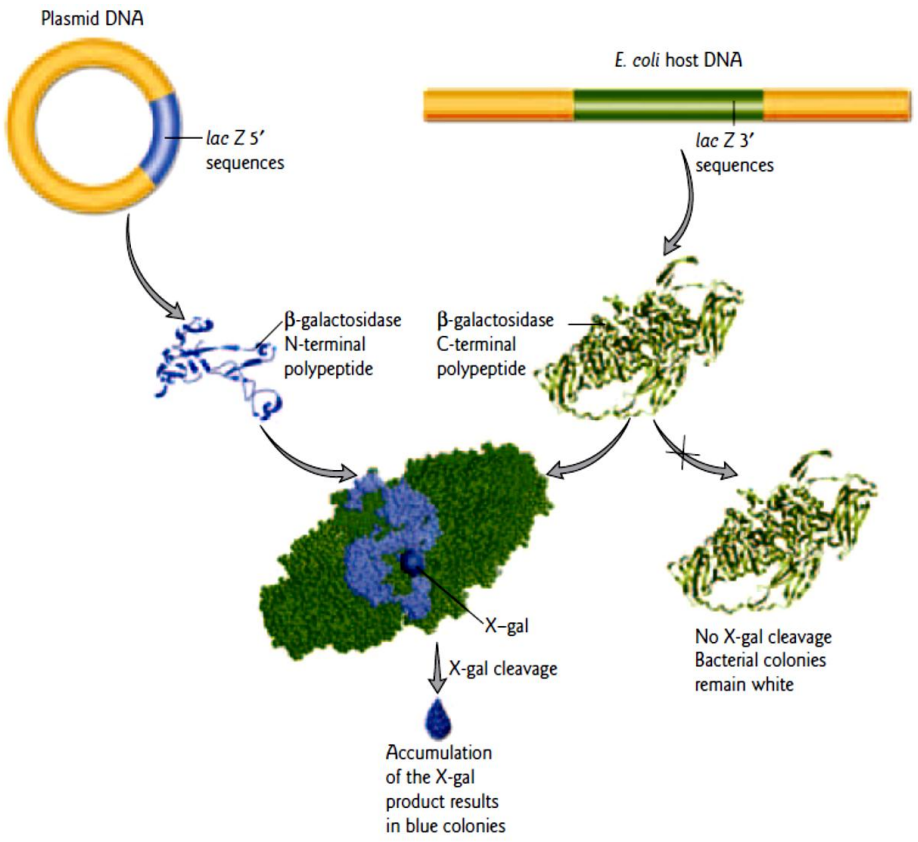



**Figure 8.7 β-Galactosidase activity can be used as an indicator of the presence of a foreign DNA insert.** Plasmids that express the N-terminal fragment of β-galactosidase (*lacZ* 5′) can be used in *E. coli* strains expressing the C-terminal fragment of the enzyme (*lacZ* 3′ sequences). The N-terminal and C-terminal fragments join and four subunits come together to form a functional tetrameric-enzyme. β-Galactosidase activity can be measured in live cells using a colorless chromogenic substrate called 5-bromo-4-chloro-3-indolyl-β-D-galactoside (X-gal). Cleavage of X-gal produces a blue-colored product that can be visualized as a blue colony on agar plates. If a foreign insert has disrupted the *lacZ* 5′ coding sequence, then only the C-terminal polypeptide will be produced in the bacterial cell. Thus, X-gal is not cleaved and bacterial colonies remain white.

## AMPLIFICATION AND PURIFICATION OF RECOMBINANT PLASMID DNA

Further screening of positive (white) colonies can be done by restriction endonuclease digest to confirm the presence and orientation of the insert (see Section 8.9). When a positive colony containing recombinant plasmid DNA is transferred aseptically to liquid growth medium, the cells will continue to multiply exponentially. Within a day or two, a culture containing trillions of identical cells can be harvested. The final step in molecular cloning is the recovery of the cloned DNA. Plasmid DNA can be purified from crude cell lysates by chromatography using silica gel or anion exchange resins that preferentially bind nucleic acids under appropriate conditions and allow for the removal of proteins and polysaccharides. The purified plasmid DNA can then be eluted and recovered by ethanol precipitation in the presence of monovalent cations. Ethanol precipitation of plasmid DNA from aqueous solutions yields a clear pellet that can be easily dissolved in an appropriate buffered solution.

## OTHER VECTOR SYSTEMS

### VIRAL VECTORS

Viral vectors, in which the gene or genes of interest are incorporated into the genome of a virus, offer many advantages for cloning and the subsequent applications of cloned genes. Because viruses infect cells with high efficiency, the cloned gene can be introduced into cells at a significantly higher frequency than by simple transformation. Some viral vectors are specialized for producing high levels of proteins encoded by the cloned genes, as exemplified by the use of insect baculovirus to express foreign proteins in a eukaryotic cell system. Other viral vectors, such as the bacterial M13-based vectors, are designed to facilitate sequencing and the generation of mutations in cloned genes. Vectors derived from retroviruses can effect the stable integration into mammalian chromosomes of cloned DNA, allowing continued expression of the gene. Viral vectors are also the vehicles of choice for gene-therapy strategies. Some examples of viral vectors used in bacteria are described next.

### PHAGE LAMBDA

Phage λ is a convenient cloning vector for several reasons. First, λ phage heads will selectively package a chromosome about 50 kb in length, and, as will be seen, this property can be used to select for λ molecules with inserts of donor DNA. The central part of the phage genome is not required for replication or packaging of λ DNA molecules in E. coli, so the central part can be cut out by using restriction enzymes and discarded. The two "arms" are ligated to restriction-enzyme-cut donor DNA. The chimeric molecules can be either introduced into E. coli directly by transformation or packaged into phage heads in vitro. In the in vitro system, DNA and phage-head components are mixed together, and infective λ phages form spontaneously. In either method, recombinant molecules with 10- to 15-kb inserts are the ones that will be most effectively packaged into phage heads, because this size of insert substitutes for the deleted central part of the phage genome and brings the total molecule size to 50 kb. Therefore the presence of a phage plaque on the bacterial lawn automatically signals the presence of recombinant phage bearing an insert. A second useful property of a phage vector is that recombinant molecules are automatically packaged into infective phage particles, which can be conveniently stored and handled experimentally.

Career Avenues

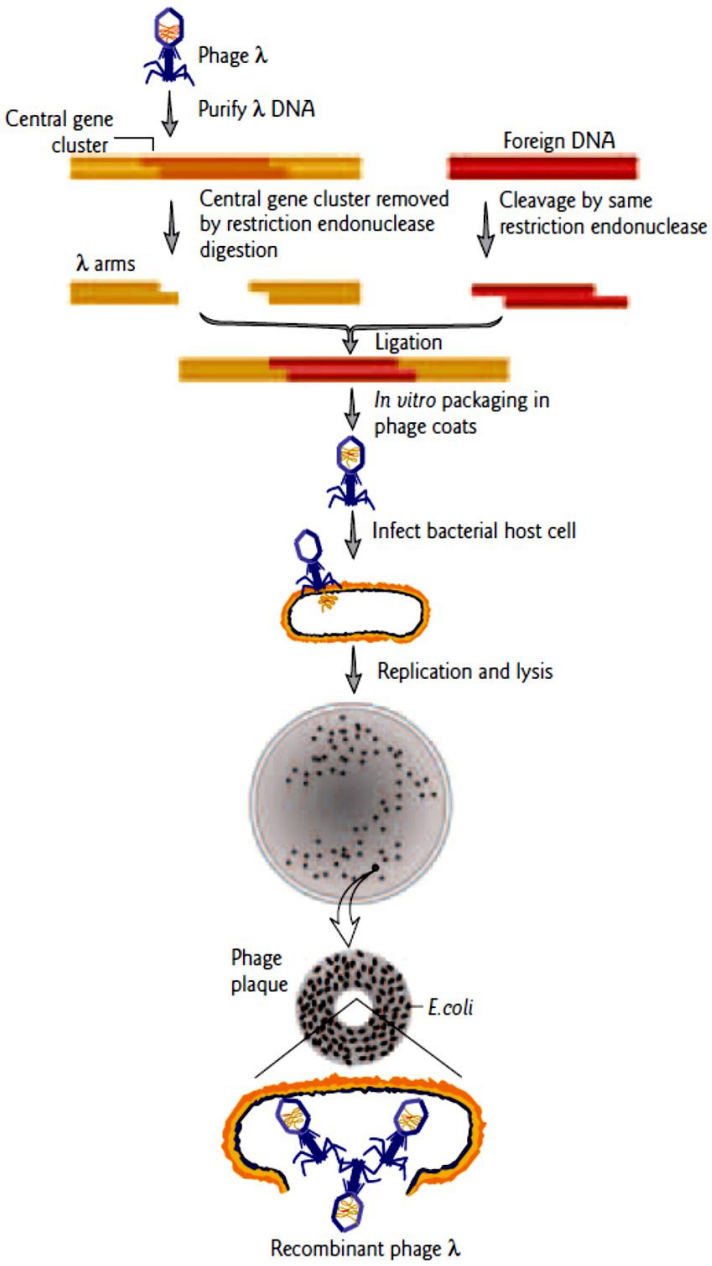



**Figure: Use of bacteriophage lambda (λ) as a cloning vector.** DNA is extracted from phage λ and the central gene cluster is removed by restriction endonuclease digestion. The foreign DNA to be cloned is cut with the same enzyme and ligated to the left and right "arms" of the phage λDNA. The recombinant DNA is then mixed with phage proteins *in vitro*. The DNA is packaged into the phage head and tail fibers are attached via a self-assembly pathway. The recombinant viral particle is then able to infect bacterial cells on an agar plate. The phage replicates its genome, including the foreign DNA insert. Recombinant phage λDNA directs the cell to make phage particles. The bacteria become filled with new phage particles, break open (lyse), and release millions of recombinant phages. The holes in the lawn of host bacteria, called plaques, are regions where phages have killed the bacteria. Each plaque represents progeny of a single recombinant phage.

## SINGLE-STRANDED PHAGES

Some phages contain only single-stranded DNA molecules. On infection of bacteria, the single infecting strand is converted into a double-stranded replicative form, which can be isolated and used for cloning. The advantage of using these phages as cloning vectors is that single-stranded DNA is the very substrate required for the Sanger DNA-sequencing technique currently in widespread use. Phage M13 is the one most widely used for this purpose.

## COSMIDS

Cosmids are vectors that are hybrids of λ phages and plasmids, and their DNA can replicate in the cell like that of a plasmid or be packaged like that of a phage. However, cosmids can carry DNA inserts about three times as large as those carried by λ itself (as large as about 45 kb). The key is that most of the λ phage structure has been deleted, but the signal sequences that promote phage-head stuffing **(cos sites)** remain. This modified structure enables phage heads to be stuffed with almost all donor DNA. Cosmid DNA can be packaged into phage particles by using the in vitro system.

## EXPRESSION VECTORS

One way of detecting a specific cloned gene is by detecting its protein product expressed in the bacterial cell. Therefore, in these cases, it is necessary to be able to express the gene in bacteria; that is, to transcribe it and translate the mRNA into protein. Most cloning vectors do not permit expression of cloned genes, but such expression is possible if special vectors are used. However, because bacteria cannot process introns, the cloned sequences must be stripped of introns. The cloned gene is inserted next to appropriate bacterial transcription and translation start signals. Some expression vectors have been designed with restriction sites located just next to a lac regulatory region. These restriction sites permit foreign DNA to be spliced into the vector for expression under the control of the lac regulatory system.

## ARTIFICIAL CHROMOSOME VECTORS

Bacterial artificial chromosomes (BACs) and yeast artificial chromosomes (YACs) are important tools for mapping and analysis of complex eukaryotic genomes. Much of the work on the Human Genome Project and other genome sequencing projects depends on the use of BACs and YACs, because they can hold greater than 300 kb of foreign DNA. BACs are constructed using the fertility factor plasmid (F factor) of *E. coli* as a starting point. The plasmid is naturally 100 kb in size and occurs at a very low copy number in the host. The engineered BAC vector is 7.4 kb (including a replication origin, cloning sites, and selectable markers) and thus can accommodate a large insert of foreign DNA. The characteristics of YAC vectors are discussed below.

Career Avenues

Immediately after the construction of the first YAC in 1983, efforts were undertaken to develop a mammalian artificial chromosome (MAC). From there on, it took 14 years until the first prototype MAC was described in 1997. Like YACs, MACs rely on the presence of centromeric sequences, sequences that can initiate DNA replication, and telomeric sequences. Their development is considered an important advance in animal biotechnology and human gene therapy for two main reasons. First, they involve autonomous replication and segregation in mammalian cells, as opposed to random integration into chromosomes (as for other vectors). Second, they can be modified for their use as expression systems of large genes, including not only the coding region but all control elements. A major drawback limiting application at this time, however, is that they are difficult to handle due to their large size and can be recovered only in small quantities. Two principal procedures exist for the generation of MACs. In one method, telomere-directed fragmentation of natural chromosomes is used. For example, a human artificial chromosome (HAC) has been derived from chromosome 21 using this method. Another method involves *de novo* assembly of cloned centromeric, telomeric, and replication origins *in vitro*.

## YEAST ARTIFICIAL CHROMOSOME (YAC) VECTORS

Yeast, although a eukaryote, is a small single cell that can be manipulated and grown in the lab much like bacteria. YAC vectors are designed to act like chromosomes. Their design would not have been possible without a detailed knowledge of the requirements for chromosome stability and replication, and genetic analysis of yeast mutants and biochemical pathways. YAC vectors include an origin of replication (autonomously replicating sequence, ARS), a centromere to ensure segregation into daughter cells, telomeres to seal the ends of the chromosomes and confer stability, and growth selectable markers in each arm. These markers allow for selection of molecules in which the arms are joined and which contain a foreign insert. For example, the yeast genes *URA3* and *TRP1* are often used as markers. Positive selection is carried out by auxotrophic complementation of a *ura3-trp1* mutant yeast strain, which requires supplementation with uracil and tryptophan to grow. *URA3* encodes an enzyme that is required for the biosynthesis of the nitrogenous base uracil (orotidine-5′-phosphate decarboxylase). *TRP1* encodes an enzyme that is required for biosynthesis of the amino acid tryptophan (phosphoribosylanthranilate isomerase). YAC vectors are maintained as a circle prior to inserting foreign DNA. After cutting with restriction endonucleases *Bam*HI and *Eco*RI, the left arm and right arm become linear, with the end sequences forming the telomeres. Foreign DNA is cleaved with *Eco*RI and the YAC arms and foreign DNA are ligated and then transferred into yeast host cells. The yeast host cells are maintained as spheroplasts (lacking yeast cell wall). Yeast cells are grown on selective nutrient regeneration plates that lack uracil and tryptophan, to select for molecules in which the arms are joined bringing together the *URA3* and *TRP1* genes.

Career Avenues

## SOURCES OF DNA FOR CLONING

The cloning that has been described so far will work for any random piece of DNA. But since the goal of many cloning experiments is to obtain a sequence of DNA that directs the production of a specific protein, we need to first consider where to obtain such DNA. Sources of DNA for cloning into vectors may be DNA fragments representing a specific gene or portion of a gene, or may be sequences of the entire genome of an organism, depending on the end goal of the researcher. Typical "inserts" include genomic DNA, cDNA, polymerase chain reaction (PCR) products and chemically synthesized oligonucleotides. When previously isolated clones are transferred into a different vector for other applications, this is called "subcloning."

Career Avenues

# cDNA AND GENOMIC DNA LIBRARY

## CONSTRUCTING DNA LIBRARIES

Vectors are used to compile a library of DNA fragments that have been isolated from the genomes of a variety of organisms. This collection of fragments can then be used to isolate specific genes and other DNA sequences of interest. DNA fragments are generated by cutting the DNA with a specific restriction endonuclease. These fragments are ligated into vector molecules, and the collection of recombinant molecules is transferred into host cells, one molecule in each cell. The total number of all DNA molecules makes up the library. This library is searched, that is screened, with a molecular probe that specifically identifies the target DNA. Once prepared the library can be perpetuated indefinitely in the host cells and is readily retrieved whenever a new probe is available to seek out a particular fragment. Two main types of libraries can be used to isolate specific DNAs: genomic and cDNA libraries.

## GENOMIC LIBRARY

A genomic library contains DNA fragments that represent the entire genome of an organism. The first step in creating a genomic library is to break the DNA into manageable size pieces (e.g. 15–20 kb for phage λ vectors), usually by partial restriction endonuclease digest. Under limiting conditions, any particular restriction site is cleaved only occasionally, so not all sites are cleaved in any particular DNA molecule. This generates a continuum of overlapping fragments. The second step is to purify fragments of optimal size by gel electrophoresis or centrifugation techniques. The final step is to insert the DNA fragments into a suitable vector. In humans, the genome size is approximately $3 \times 10^9$ bp. With an average insert size of 20 kb, the number of random fragments to ensure with high probability (95–99%) that every sequence is represented is approximately 106 clones for humans. The maths actually works out to $1.5 \times 10^5$ (i.e. ($3 \times 10^9$ bp)/ ($2 \times 10^4$ bp)) but more clones are needed in practice, since insertion is random. Bacteriophage λ or cosmid vectors are typically used for genomic libraries. Since a larger insert size can be accommodated by these vectors compared with plasmids, there is a greater chance of cloning a gene sequence with both the coding sequence and the regulatory elements in a single clone.

## CDNA LIBRARY

The principle behind cDNA cloning is that an mRNA population isolated from a specific tissue, cell type, or developmental stage (e.g. embryo mRNA) should contain mRNAs specific for any protein expressed in that cell type or during that stage, along with "housekeeping" mRNAs that encode essential proteins such as the ribosomal proteins, and other mRNAs common to many cell types or stages of development. Thus, if mRNA can be isolated, a small subset of all the genes in a genome can be studied. mRNA cannot be cloned directly, but a cDNA copy of the mRNA can be cloned. Because a cDNA library is derived from mRNA, the library contains the

Career Avenues

coding region of expressed genes only, with no introns or regulatory regions. This latter point becomes important for applications of recombinant DNA technology to the production of transgenic animals and for human gene therapy.

## CDNA SYSNTHESIS

### FIRST STRAND SYNTHESIS OF CDNA:

A number of strategies can be used to synthesize cDNA from purified mRNA. One strategy is as follows. In brief, cDNA is synthesized by the action of reverse transcriptase and DNA polymerase. The reverse transcriptase catalyzes the synthesis of a single-stranded DNA from the mRNA template. Like a regular DNA polymerase, reverse transcriptase also needs a primer to get started. A poly(dT) primer is added to provide a free 3′-OH end that can be used for extension by reverse transcriptase in the presence of deoxynucleoside triphosphates (dNTPs). Usually a viral reverse transcriptase is employed such as one from avian myeloblastosis virus (AMV). The reverse transcriptase adds dNTPs from 5′ to 3′ by complementary base pairing. This is called first strand synthesis. The mRNA is then degraded with a ribonuclease or an alkaline solution.

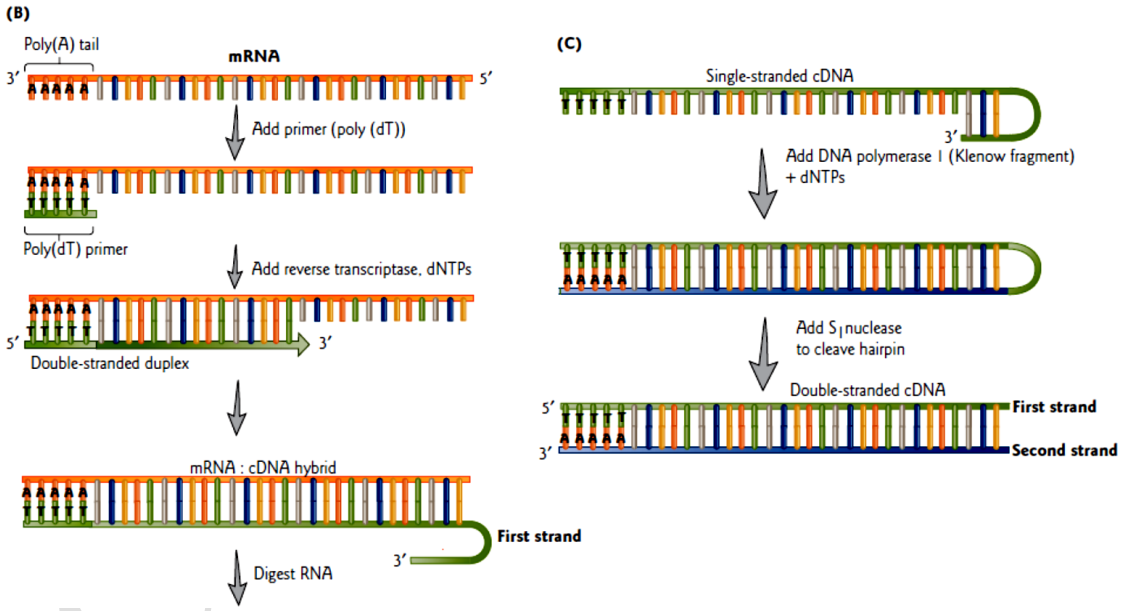



**Figure: cDNA Synthesis:** (B) First strand synthesis. Synthesis of the first strand of cDNA is carried out using the enzyme reverse transcriptase and a poly(dT) primer in the presence of dNTPs. An mRNA–cDNA hybrid is produced and the mRNA is then digested with an alkaline solution or the enzyme ribonuclease. (C) Second strand synthesis. Synthesis of double-stranded cDNA uses a self-priming method. The Klenow fragment of DNA polymerase I catalyzes synthesis of the second strand, using the natural hairpin of the first strand as a primer. The hairpin is cleaved with a single-strand DNA nuclease (S1 nuclease). The end result is a collection of double-strand cDNAs that correspond to the sequences of the many different mRNAs extracted from the cell.

## SECOND STRAND SYNTHESIS OF CDNA

For most applications, including cloning of cDNAs, double stranded DNA is required. The second DNA strand is generated by the Klenow fragment of DNA polymerase I from *E. coli*. The 5′→3′exonuclease activity of DNA polymerase I from *E. coli* makes it unsuitable for many applications. However, this enzymatic activity can be readily removed from the holoenzyme by exposure to a protease. The large or Klenow fragment of DNA polymerase I generated by proteolysis has 5′→3′ polymerase and 3′→5′ exonuclease (proofreading) activity, and is widely used in molecular biology. Commercially available Klenow fragments are usually produced by expression in bacteria from a truncated form of the DNA polymerase I gene.

There is a tendency for the reverse transcriptase enzyme used in first strand synthesis to loop back on itself and start to make another complementary strand. This hairpin forms a natural primer for DNA polymerase and a second strand of DNA is generated. S1 nuclease (from *Aspergillus oryzae*) is then added to cleave the single-stranded DNA hairpin. Double-strand DNA linkers with ends that are complementary to an appropriate cloning vector are added to the double-strand DNA molecule before ligation into the cloning vector. The end result is a double-stranded cDNA in which the second strand corresponds to the sequence of the mRNA, thus representing the coding strand of the gene. The sequences that appear in the literature are the 5′→3′ sequences of the second strand cDNA. Sequences corresponding to introns and to promoters and all regions upstream of the transcriptional start site are not represented in cDNAs. The library created from all the cDNAs derived from the mRNAs in the specific cell type forms the cDNA library of cDNA clones.

Career Avenues

# PROBES

Searching for a specific cloned DNA sequence in a library is called library screening. One of the key elements required to identify a gene during library screening is the probe. The term probe generally refers to a nucleic acid (usually DNA) that has the same or a similar sequence to that of a specific gene or DNA sequence of interest, such that the denatured probe and target DNA can hybridize when they are renatured together. The probe not only must have the same or a similar sequence to the gene of interest but the researcher must also be able to detect its hybridization. Thus, the probe is labeled; that is, it is chemically modified in some way which allows it, and hence anything it hybridizes to, to be detected. Specific enzymes are used that can add labeled nucleotides in a variety of ways. Typically the probe is made radioactive and added to a solution. Filters containing immobilized clones are then bathed in the solution. The principle behind this step is that the probe will bind to any clone containing sequences similar to those found on the probe. This binding step is called hybridization. In some cases a library is screened with a protein. For example, when a cDNA library is being screened an antibody can be used to identify the protein that is being expressed by the insert of the clone. In this case, the library is said to be "incubated" with the antibody probe, not hybridized.

Hybridization can occur between DNA and DNA, DNA and RNA, and RNA and RNA. There are three major types of probe:

- oligonucleotide probes, which are synthesized chemically and end-labeled;
- DNA probes, which are cloned DNAs and may either be end-labeled or internally labeled during *in vitro* replication; and
- RNA probes (riboprobes), which are internally labeled during *in vitro* transcription from cloned DNA templates. RNA probes and oligonucleotide probes are generally single-stranded. DNA may be labeled as a double-stranded or single-stranded molecule, but it is only useful as a probe when singlestranded and therefore must be denatured before use.

Oligonucleotide, cloned DNA, and RNA probes are of two major types: heterologous and homologous.

## HETEROLOGOUS PROBES

A heterologous probe is a probe that is similar to, but not exactly the same as, the nucleic acid sequence of interest. If the gene being sought is known to have a similar nucleotide sequence to a second gene that has already been cloned, then it is possible to use this known sequence as a probe. For example, a mouse probe could be used to search a human genomic library.

Career Avenues

## HOMOLOGOUS PROBES

A homologous probe is a probe that is exactly complementary to the nucleic acid sequence of interest. Homologous probes can be designed and constructed in a number of different ways. Examples include degenerate probes, expressed sequence tag (EST) based probes, and cDNA probes that are used to locate a genomic clone.

**Some radioisotopes commonly used in molecular biology research**

| Radioisotope | Symbol | Labeled macromolecule | Half-life* | Application |
|---|---|---|---|---|
| Tritium (hydrogen-3) | $^{3}H$ | Nitrogenous bases (e.g. $^{3}H$-uridine, $^{3}H$-thymidine)<br>$^{3}H$-dNTPs, $^{3}H$-NTPs | 12.28 years | RNA and DNA labeling *in vivo*<br>Probes for *in situ* hybridization |
| Carbon-14 | $^{14}C$ | $^{14}C$-chloramphenicol | 5730 years | CAT assays |
| Phosphorus-32 | $^{32}P$ | $^{32}P$-NTPs<br>$^{32}P$-dNTPs | 14.29 days | Hybridization probes |
| Sulfur-35 | $^{35}S$ | Amino acids ($^{35}S$-cysteine, $^{35}S$-methionine) | 87.4 days | Protein labeling (*in vivo* and *in vitro*) |
| Iodine-125 | $^{125}I$ | N/A | 60.14 days | Antibody labeling |

## UNIQUE EST-BASED PROBES

The use of EST-based probes is a newer method than making degenerate probes. Although EST-based probes are no longer frequently used for organisms in which the whole genome sequence is available, they are still useful for organisms where only limited sequence information is available. ESTs are partial cDNA sequences of about 200–400 bp (because they represent just a short portion of the cDNA, they are called "tags"). This method uses cDNA sequence data and identifies a single oligonucleotide rather than a degenerate mixture. A computer program applies the genetic code to translate an EST into a partial amino acid sequence. If a match is found with the protein under study, the EST provides the unique DNA sequence of that portion of cDNA. A probe can then be synthesized and used to screen a library for the entire cDNA (or genomic) clone.

## USING AN IDENTIFIED CDNA TO LOCATE A GENOMIC CLONE

Since a cDNA contains only the coding region of a gene, researchers often need to isolate a genomic clone for analysis of regulatory regions, introns, etc. Use of an identified cDNA to locate a genomic clone provides a highly specific probe for the gene of interest.

Career Avenues

# TRANSPOSONS AND GENE TARGETING

## GENE TAGGING

The use of transposable elements or transgenes for insertional mutagenesis is advantageous over traditional mutagenesis methods, because the interrupted locus becomes 'tagged' with a unique sequence. Cloning genes tagged with endogenous transposons can sometimes be complicated if there are many copies of the element in the genome, but the use of heterologous transposons or transgenes can provide a truly unique sequence tag that identifies the interrupted gene. A genomic library can therefore be generated from the mutant, and the interrupted gene can be identified using hybridization or the polymerase chain reaction (PCR) to identify clones containing the tag. Such clones will also contain flanking DNA from the interrupted genomic locus.

Compared with chromosome walking and similar techniques, tagging is a simple procedure that homes in directly on the mutant gene. Several PCR-based techniques can be used to directly amplify genomic DNA flanking a transposon or transgene tag, thus avoiding the necessity for library construction in phage-λor cosmid vectors. Genomic DNA from the mutant is digested with a restriction enzyme for which there is no site in the insertional tag. The DNA is thus divided into fragments, one of which will contain the entire insertion sequence flanked by genomic DNA tails. The tails, however, have compatible sticky ends and in a suitably dilute ligation reaction can be self-ligated to generate a circular genomic fragment. PCR primers annealing at the edges of the transposon tag but facing *outwards* can then be used to amplify the flanking genomic sequences. Such PCR strategies have been used to generate fragment libraries of flanking sequences for the rapid identification of genes associated with particular phenotypes. If appropriate, these products can be used to screen traditional genomic or cDNA libraries to isolate the uninterrupted clone.

An even greater simplification of this procedure is possible if the origin of replication from a bacterial plasmid and an antibiotic-resistance gene are included in the tag. In this case, self-ligation of the construct as above generates an autonomously replicating plasmid containing genomic DNA. If whole preparations of genomic DNA are used to transform competent bacterial cells, only the tag-derived plasmid will replicate, while other genomic circles will be rapidly diluted from the culture. Furthermore, only bacteria carrying the antibiotic-resistance marker will survive selection, allowing pure plasmid DNA corresponding to the flanking regions of the genomic insert to be automatically prepared.

Both endogenous transposable elements and transgene insertions have therefore been used to isolate genes by tagging in a variety of species. In *D. melanogaster*, P elements have been widely used because of the high transposition frequency and the availability of strains that lack P elements, allowing lines containing individual insertions of interest to be derived by back-crossing. *Tc*1 elements have been used for transposon tagging in *C. elegans*, *Ac–Ds* elements

Career Avenues

have been widely used in maize and *Tam* elements have been used in the snapdragon (*Antirrhinum majus*).

*Tam* elements are particularly useful because the transposition frequency can be increased more than 800-fold if the temperature at which the plants are raised is reduced to 15°C. Also, it is notable that approximately 5% of all naturally occurring recessive mutations in the mouse are caused by a particular family of retroviruses. The cloning of several mouse genes has been facilitated by virtue of their linkage to a proviral sequence The transposition of *Drosophila* P elements appears to be restricted to drosophiloid insects, but other transposable elements are more promiscuous. The *Ac–Ds* elements of maize have been shown to function in a wide range of plants – dicots as well as monocots – and have been extensively used as insertional mutagens and gene tags in heterologous species. T-DNA can also be used as a tag following transfer by *Agrobacterium tumefaciens*. T-DNA mutagenesis and tagging has been used in *Arabidopsis* and has recently been extended to other plant systems, such as rice.

Career Avenues

# NUCLEIC ACID LABELING

There are a variety of methods for labeling RNA and DNA. The choice of method depends on the application. Probes of the highest specific activity (proportion of incorporated label per mass of probes) are generated using internal labeling, where many labeled nucleotides are incorporated uniformly during DNA or RNA synthesis *in vitro*. End-labeling involves either adding a labeled nucleotide to the 3′-hydroxyl end of a DNA strand or exchanging the unlabeled 5′-phosphate group for a labeled phosphate. When deciding on a labelling method, one of the first questions to ask is whether internal labeling or end-labeling of the nucleic acid is desirable. Internal labeling provides maximal labeling and is often used for probes to screen libraries or for Southern blots. End-labeling is used when precise definition of one end of the DNA (5′ or 3′) is required.

An example of this is labeling a DNA fragment for use in DNase I footprinting, where the researcher wants to know the orientation of the fragments; that is, is a protein binding site near the 5′ or 3′ end of the DNA sequence?

Common methods of uniform labeling involve DNA or RNA synthesis reactions; for example, random primed labelling and synthesis of riboprobes. Some methods for end-labeling DNA fragments involve DNA synthesis reactions (e.g. Klenow fill-in), while oligonucleotides are generally endlabeled by other enzyme-mediated reactions.

## RANDOM PRIMED LABELLING

Random primed labeling is a method of incorporating radioactive nucleotides along the length of a fragment of DNA. The DNA is denatured and random oligonucleotides are annealed to both strands. Each batch of random oligonucleotides contains all possible sequences (for hexamers, which are most commonly employed, this would be 4096 different oligonucleotides), so any DNA template can be used with this method. The odds are that some of these primers will anneal to the DNA of interest. The oligonucleotide primers provide the required free 3′- hydroxyl group for the initiation of DNA synthesis. A Klenow fragment of *E. coli* DNA polymerase I is then used to extend the oligonucleotides, using three unlabeled (“cold”) nucleotides and one radioactively labeled (“hot”) nucleotide (or a nonradioactively labeled nucleotide) provided in the reaction mixture, to produce a uniformly labeled double stranded probe. Subsequently, the double-stranded probe is denatured and added to a hybridization solution.

## IN VITRO TRANSCRIPTION

RNA can be labeled by *in vitro* transcription from a DNA template. The DNA template (often a cDNA) is cloned into a special plasmid vector so that it can be transcribed under the control of a promoter specific for recognition by a bacteriophage RNA polymerase, typically SP6 RNA

Career Avenues

polymerase, T7 RNA polymerase, or T3 RNA polymerase. The transcription reaction is carried out *in vitro* by the addition of all four NTPs, with one or more labeled, and the appropriate phage RNA polymerase. Labeled RNA can be used for tracking the movement and localization of RNA transcripts in cells, analyzing RNA processing pathways, and as hybridization probes. RNA probes (riboprobes) may be either complementary to the sense or antisense strand of DNA depending on the purpose.

## KLENOW FILL-IN

A "fill-in" reaction is used to generate blunt ends on fragments created by cleavage with restriction endonucleases that leave 5′ single-stranded overhangs. The Klenow fragment of *E. coli* DNA polymerase I is used to fill in the gaps from 5′ to 3′, in the presence of dNTPs, including one labeled dNTP. The result is a doublestranded DNA with the 3′□ends labeled.

Career Avenues

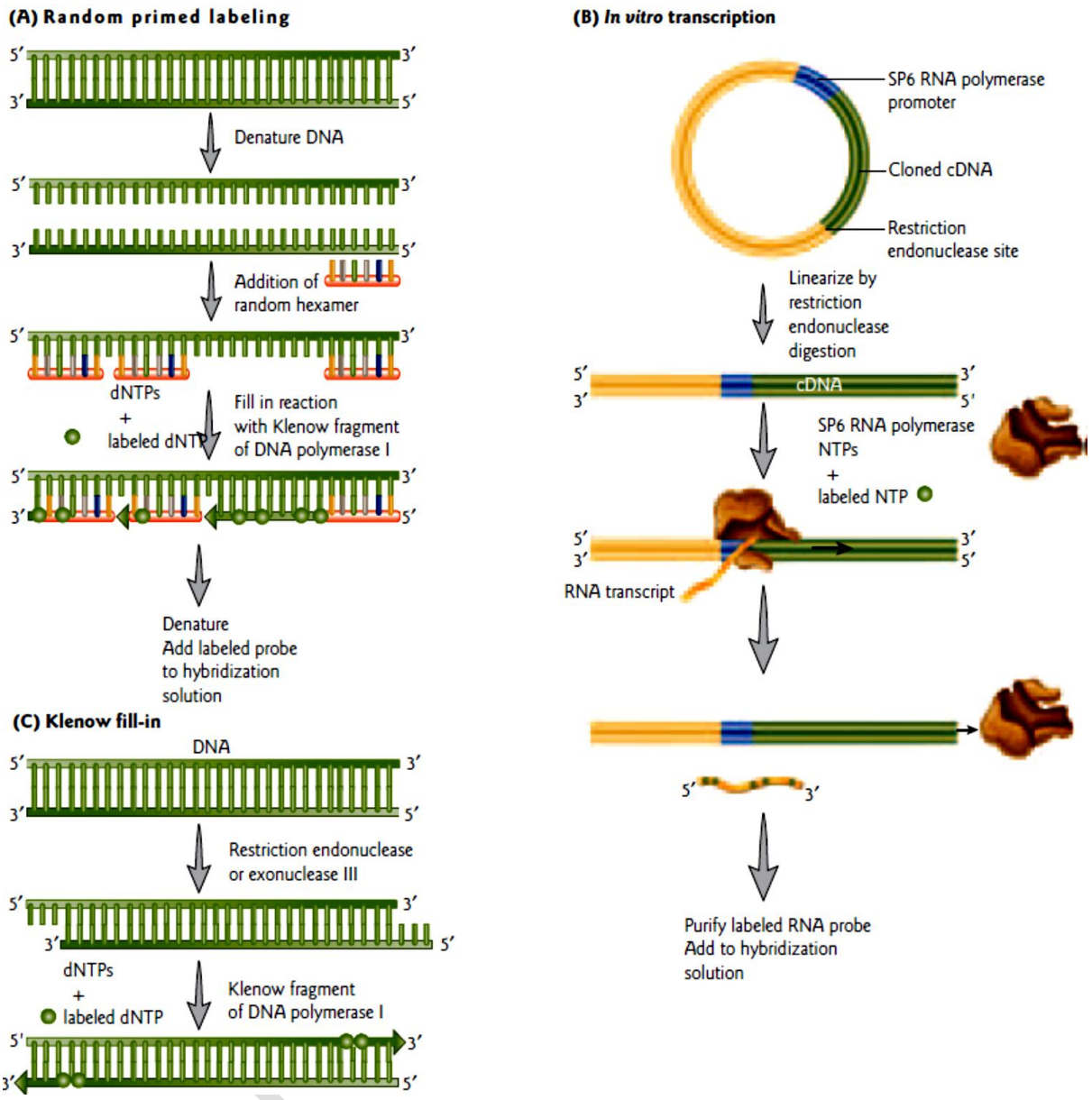



**Figure: Some methods for labeling nucleic acids.** Two methods of uniform labeling and one method of end-labeling are depicted that involve nucleic acid synthesis reactions. (A) In random primed labeling, the template DNA is denatured and random hexamer primers are annealed to both strands. Only one strand is shown for simplicity. The primers provide the 3′-OH for the initiation of DNA synthesis upon addition of the Klenow fragment of DNA polymerase I, and unlabeled and labeled dNTPs. The resulting labeled double-stranded DNA probe is denatured and added to a hybridization solution. (B) *In vitro* transcription generates a labeled RNA from a DNA template. The DNA template (cDNA) is cloned in a plasmid vector containing a promoter for bacteriophage SP6 RNA polymerase. The transcription reaction is carried out by the addition of all labeled and unlabeled NTPs and SP6 RNA polymerase. The labeled RNA probe can then be purified and added to a hybridization solution. (C) Klenow fill-in is used to create labeled blunt ends on fragments created by cleavage with a restriction endonuclease that leaves a 5′

overhang. The reaction is conceptually identical to the one described in (A), but with a long primer and a very short segment of single-stranded template.

## LABELING OF NUCLEIC ACIDS

| Labeling method | Type of labeling | Enzyme | Example of application |
|---|---|---|---|
| **Random priming** | Uniform | Klenow DNA polymerase | Hybridization probes |
| ***In vitro* transcription** | Uniform | Phage SP6, T7, or T3 RNA polymerase | Hybridization probes, tracking RNA localization |
| **Klenow fill-in** | 3′ end-labeling | Klenow DNA polymerase | DNase I footprinting |
| **Oligonucleotides** | 5′ end-labeling<br>3′ end-labeling | T4 polynucleotide kinase<br>Terminal transferase | Hybridization probes, EMSA |

Career Avenues

# LIBRARY SCREENING AND RFLP

Nowadays, because of the wealth of genomic sequence data available for many organisms, a DNA sequence of interest is more likely to be isolated by polymerase chain reaction (PCR) than by a library screen. DNA cloning and PCR both amplify tiny samples of DNA into large quantities by repeated rounds of DNA duplication, either carried out by cycles of cell division in a host or cycles of DNA synthesis *in vitro*. In PCR, the pair of oligonucleotide primers limits the amplification process to the particular DNA sequence of interest from the beginning. In contrast, once prepared, a DNA library can be perpetuated indefinitely in host cells, and can be readily retrieved whenever the researcher wants to seek out some particular fragment. Assuming a probe is available for the cloned sequence of interest, the library can be screened using the principles of hybridization. Complementary base pairing of single-stranded nucleic acids underlies some of the most important biological processes: replication, recombination, DNA repair, transcription, and translation. Library screening exploits this fundamental ability of double-stranded nucleic acids to undergo denaturation or melting (separation into single strands) and for complementary single strands to spontaneously anneal to a labeled nucleic acid probe to form a hybrid duplex (heteroduplex).

The power of the technique is that the labeled nucleic acid probe can detect a complementary molecule in a complex mixture with exquisite sensitivity and specificity. Like many procedures in molecular biology, the process of library screening sounds simple, but in practice it can be labor intensive, and may result in "false positives" if the hybridization conditions are not stringent enough. A similar protocol would be employed for screening a phage library, but would involve plaque hybridization instead of colony hybridization.

## TRANSFER OF COLONIES TO A DNA-BINDING MEMBRANE

Bacterial colonies with recombinant vectors containing inserts representing the entire library are grown on nutrient agar plates, forming hundreds or thousands of colonies. The colonies (members of the library) are transferred to nitrocellulose or nylon membranes by gently pressing to make a replica. Bacteria attach to the membrane and the cells are then lysed and the DNA purified by treatment with alkali and proteases. The DNA is denatured to make it single stranded, and fixed to the membrane either by heat treatment or ultraviolet irradiation. The DNA is covalently bound by its sugar–phosphate backbone and the unpaired bases are exposed for complementary base pairing.

## COLONY HYBRIDIZATION

In the next step of library screening, a radioactively labeled, single-stranded DNA probe is applied. The hybridization step is performed at a nonstringent temperature that ensures the probe will bind to any clone containing a similar sequence. At the same time, some nonspecific

Career Avenues

hybridization will occur because some of the clones will contain limited, but not significant, similarity to the probe. A series of washes are performed at a stringent temperature that is high enough to remove the probe from all clones to which it has bound in a nonspecific manner. Heteroduplex stability is influenced by the number of hydrogen bonds between the bases and base stacking by hydrophobic interactions that hold the two single strands together. The number of hydrogen bonds is determined by various properties of the heteroduplex, including the length of the duplex, its GC content, and the degree of mismatch between the probe and the complementary target DNA sequence. The shorter the duplex, the lower the GC content, and the more mismatches there are, the lower the melting temperature (*T*m) will be because there are fewer hydrogen bonds and base-stacking interactions to disrupt. The appropriate hybridization temperature is calculated according to the GC content and the percent homology of probe to target according to the following equation:

$$T_m = 49.82 + 0.41(\%\ G + C) - (600/l)$$

where, *l* is the length of the hybrid in base pairs.

It is important that the temperature is not so high that it removes the probe from clones that contain sequences that are similar (only a few mismatches) or identical to the probe itself (exact complementarity). Therefore, consideration about the source of the probe (homologous or heterologous) determines the temperature at which the high stringency washes are performed.

## DETECTION OF POSITIVE COLONIES

In the final phase of library screening, an X-ray film is applied to the membrane and is exposed by any remaining specifically bound radioactive probe. The resulting autoradiogram has a dark spot on the developed film where DNA–DNA hybrids have formed, by virtue of sequence complementarity to the radiolabeled probe. If the gene is large, it may be fragmented. Various fragments in different clones may need to be identified by finding overlapping fragments and reconstructing the order. The original plate is used to pick bacterial cells with recombinant plasmids that hybridized to the probe. Cells are transferred to medium for growth and further analysis.

## EXPRESSION LIBRARIES

Expression libraries are made with a cloning vector that contains the required regulatory elements for gene expression, such as the promoter region (see Section 10.3). In an *E. coli* expression vector (plasmid or bacteriophage), an *E. coli* promoter is placed next to a unique restriction site where DNA can be inserted. When a foreign cDNA is cloned into an expression vector in the correct reading frame, the coding region is transcribed and translated in the *E. coli* host. Expression libraries are useful for identifying a clone containing the cDNA of interest when an antibody to the protein encoded by that gene or cDNA is available. Binding of a radioactively

Career Avenues

labeled antibody, using a technique similar to nucleic acid hybridization, can be used to identify a specific protein made by one of the clones of the expression library.

## RESTRICTION MAPPING

Once the clone of interest has been isolated, the first stage of analysis is often the creation of a restriction map. Restriction mapping provides a compilation of the number, order, and distance between restriction endonuclease cutting sites along a cloned DNA fragment. In addition, restriction mapping plays an important role in characterizing DNA, mapping genes, and diagnostic tests for genetic diseases.

To make a restriction map, a cloned DNA fragment is cut with restriction endonucleases and loaded on to an agarose gel for electrophoresis. The lengths of the DNA fragments can be determined by comparing their position in the gel to reference DNAs of known lengths in the gel. A DNA fragment migrates a distance that is inversely proportional to the logarithm of the fragment length in base pairs over a limited range in the gel. Thus, agarose gel electrophoresis allows the restriction fragment lengths to be determined. The pattern of cutting in single and double digests indicates what the relationship is between the two sites.

Assume, for example, that you have attempted to subclone a cDNA into a plasmid vector. You have obtained a positive clone by blue-white screening. Because you want to use the recombinant plasmid DNA for the preparation of an antisense RNA probe, you need to check that the orientation of the insert in the plasmid vector is correct. According to the plasmid map shown in Fig. 8.13, if the insert is in the desired orientation, the fragment sizes generated by a double digest with *Eco*RI and *Hin*dIII will be 4.5 and 1.3 kb, respectively. If the insert is in the opposite orientation, the order of restriction sites will be reversed and the fragment sizes will be 3.5 and 2.3 kb. The results of restriction endonuclease digests show that the insert is in the correct orientation, and that you have successfully subcloned a template for *in vitro* riboprobe preparation.

Career Avenues

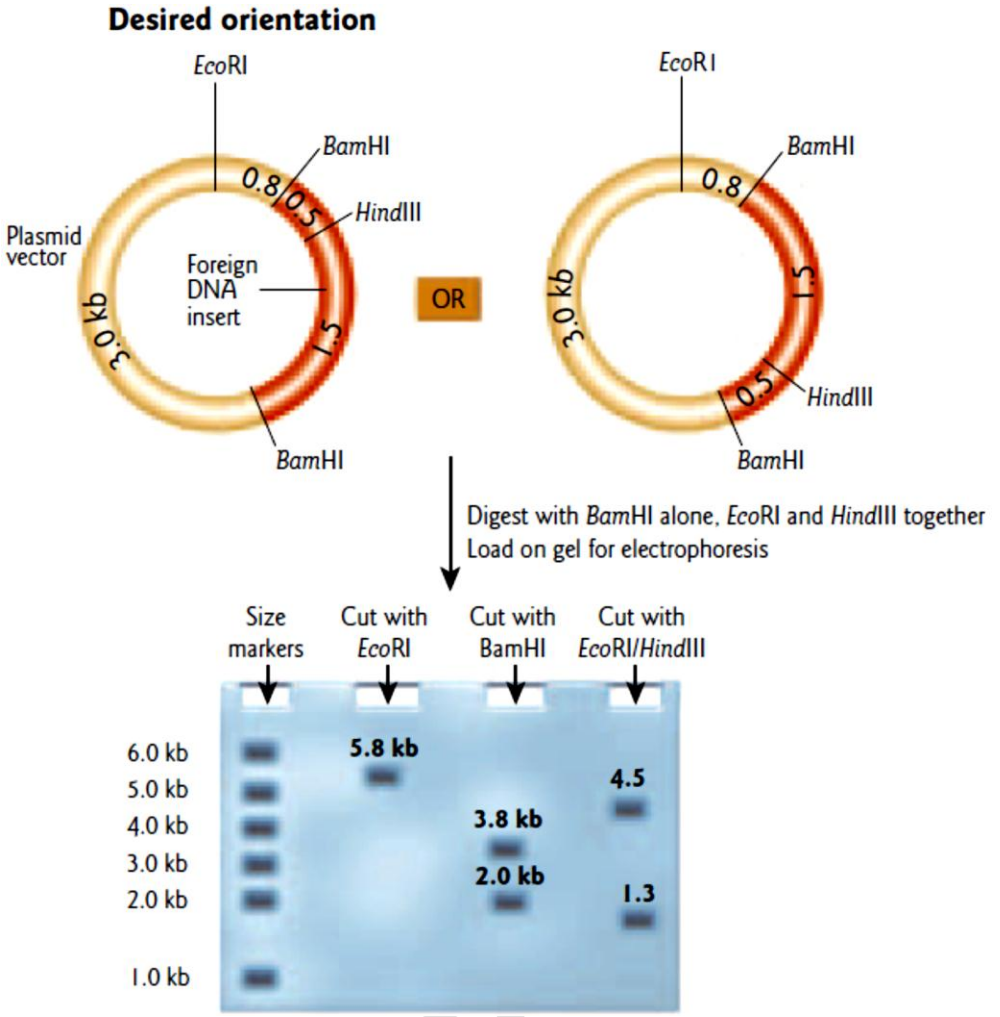



**Figure: Analysis of recombinant DNA by restriction endonuclease digestion.** Assume a 2.0 kb foreign DNA insert has been successfully ligated into the *Bam*HI site of a 3.8 kb plasmid vector. However, the orientation of the insert is unknown. Samples of the recombinant plasmid are digested with restriction endonucleases: one sample is digested with *Eco*RI, one with *Bam*HI, and one with both *Eco*RI and *Hin*dIII. The resulting fragments are separated by agarose gel electrophoresis. The sizes of the separated fragments can be measured by comparison with molecular weight standards in an adjacent lane. *Eco*RI linearizes the 5.8 kb plasmid which appears as a single band on the gel. *Bam*HI generates two fragments of 3.8 and 2.0 kb in size, representing the vector and the insert, respectively. Digestion with *Eco*RI and *Hin*dIII generates two fragments of 4.5 and 1.3 kb. These data indicate that the foreign DNA has been inserted in the desired orientation. If the DNA had been inserted in the opposite orientation, a double digest with *Eco*RI and *Hin*dIII would have generated fragment sizes of 3.5 and 2.3 kb.

Career Avenues

## RESTRICTION FRAGMENT LENGTH POLYMORPHISM (RFLP)

In 1980, Mark Skolnick, Ray White, David Botstein, and Ronald Davis created a restriction fragment length polymorphism (RFLP, pronounced "rif-lip") marker map of the human genome. A RFLP is defined by the existence of alternative alleles associated with restriction fragments that differ in size from each other. RFLPs are visualized by digesting DNA from different individuals with restriction endonucleases, followed by gel electrophoresis to separate fragments according to size, then Southern blotting, and hybridization to a labeled probe that identifies the locus under investigation. A RFLP is demonstrated whenever the Southern blot pattern obtained with one individual is different from the one obtained with another individual. These variable regions do not necessarily occur in genes, and the function of most of those in the human genome is unknown. An exception is a RFLP that can be used to diagnose sickle cell anemia. In individuals with sickle cell anemia, a point mutation in the β-globin gene has destroyed the recognition site for the restriction endonuclease *Mst*II. This mutation can be distinguished by the presence of a larger restriction fragment on a Southern blot in an affected individual, compared with a shorter fragment in a normal individual.

## RFLPS CAN SERVE AS MARKERS OF GENETIC DISEASES

By carefully examining the DNA of members of families that carry genetic diseases, it has been possible to find forms of particular RFLPs that tend to be inherited with particular diseases. The simplest RFLPs are those caused by single base pair substitutions. However, RFLPs can also be generated by the insertion of genetic material such as transposable elements, or by tandem duplications, deletions, translocations, or other chromosomal rearrangements. In linkage analysis, families in which individuals are at risk for a genetic disease are identified (i.e. both parents are heterozygous for an autosomal recessive mutation associated with a particular disease). DNA samples from various family members are then analyzed to determine the frequency with which specific RFLP markers segregate with the mutant allele causing the disease. This frequency is a measure of the distance between the markers and the mutation-defined locus. Generally, the fragment size differences occur not because a restriction site was created or disrupted by the diseased state itself, but rather because the nucleotide sequence differences just happen to be near the gene involved. A particular form of a polymorphism that is close to a diseased gene tends to stay with that gene during crossing-over (recombination) in meiosis. Relatively large segments of chromosomes are involved in crossing-over, so markers close together on a given chromosome are more likely to be transmitted together (not separated during recombination) than those that are far apart. Linkage thus refers to the likelihood of having one marker transmitted with another through meiosis. Markers that are transmitted together frequently are said to be closely linked. Thus RFLPs can serve as markers of disease, even when the RFLP is not within the disease gene. RFLPs have been useful for detecting such genetic diseases as cystic fibrosis, Huntington's disease, and hemophilia.

RFLPs were the predominant form of DNA variation used for linkage analysis until the advent of PCR. The main advantage of RFLP analysis over PCR-based protocols is that no prior sequence

Career Avenues

information or oligonucleotide synthesis is required. However, when a PCR assay for typing a particular locus is developed, it is generally preferable to RFLP analysis. In some cases, a combined approach of PCR and RFLP is used for analysis.

Career Avenues

# DNA SEQUENCING

Until 1977, determining the sequence of bases in DNA was a labor-intensive process that could be applied only to very short sequences, such as the template region for tRNA. With the development of techniques for rapid, large-scale DNA sequencing, today molecular biologists determine the order of bases in DNA as a matter of course. DNA sequencing is used to provide the ultimate characterization once a gene has been cloned or amplified by PCR. Although a sequence on its own is of limited value, it is the necessary stepping-stone to more informative analyses of the cloned gene. DNA sequencing is used to identify genes, determine the sequence of promoters and other regulatory DNA elements that control expression, reveal the fine structure of genes and other DNA sequences, confirm the DNA sequence of cDNA and other DNA synthesized *in vitro* (for example, after *in vitro* mutagenesis to confirm the mutation), and help deduce the amino acid sequence of a gene or cDNA from the DNA sequence. With the advent of automated DNA sequencing technology, large genome sequencing projects are yielding information about the evolution of genomes, the location of coding regions, regulatory elements, and other sequences, and the presence of mutations that give rise to genetic diseases.

## DNA SEQUENCING BY THE SANGER "DIDEOXY" DNA METHOD

In 1977, Frederick Sanger, Allan Maxam, and Walter Gilbert pioneered DNA sequencing. The Maxam and Gilbert sequencing method uses a chemical method that involves selective degradation of bases. The most widely used method for DNA sequencing is the Sanger or "dideoxy" method, which is, in essence, a DNA synthesis reaction. In this method, single-stranded DNA is mixed with a radioactively labeled primer to provide the 3′-OH required for DNA polymerase to initiate DNA synthesis. The primer is usually complementary to a region of the vector just outside the multiple cloning sites. The sample is then split into four aliquots, each containing DNA polymerase, four dNTPs (at high concentration), and a low concentration of a replication terminator. The replication terminators are dideoxynucleoside triphosphates (ddNTPs) that are missing the 3′-OH. Because they lack the 3′-OH, they cannot form a phosphodiester bond with another nucleotide. Thus, each reaction proceeds until a replication-terminating nucleotide is added, and each of the four sequencing reactions produces a series of single-stranded DNA molecules, each one base longer than the last. The polymerase of choice for DNA sequencing is phage T7 DNA polymerase (called "Sequenase"). The sequencing mixtures are loaded into separate lanes of a denaturing polyacrylamide gel and electrophoresis is used to separate the DNA fragments. Autoradiography is used to detect a ladder of radioactive bands. The radioactive label (primer) is at the 5′end of each newly synthesized DNA molecule. Thus, the smallest fragment at the bottom of the gel represents the 5′ end of the DNA. Reading the sequence of bases from the bottom up (5′→3′) gives the sequence of the DNA molecule synthesized in the sequencing reaction. The sequence of the original strand of DNA is complementary to the sequence read from the gel (3′→5′). This method can separate DNA of

Career Avenues

approximately 500 nt. For longer sequences, overlapping fragments are sequenced. The technique is very laborious and the sequences have to be read by hand.

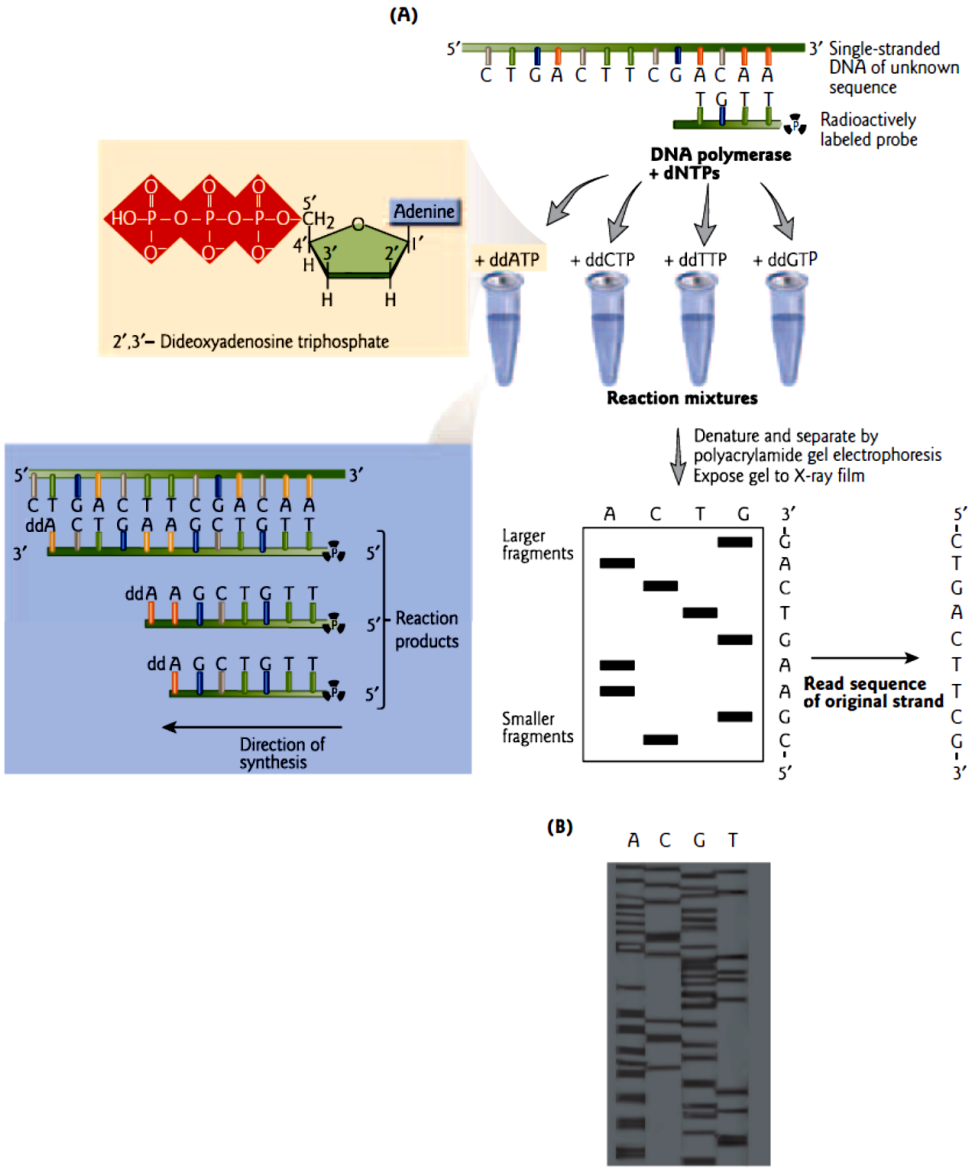



**Figure: Sanger "dideoxy" DNA sequencing.** (A) Four DNA synthesis reaction mixes are set up using template DNA, a labeled primer, DNA polymerase, and a mixture of dNTPs and one each of the four dideoxy NTPs (ddNTPs). The direction of synthesis is from 5′ to 3′. The radioactive products of each reaction mixture are separated by polyacrylamide gel electrophoresis and located by exposing the gel to X-ray film. The nucleotide sequence of the newly synthesized DNA is read directly from the autoradiogram in the 5′→3′direction, beginning at the bottom of autoradiogram. The sequence in the original template strand is its complement (3′→5′). (Inset) The random incorporation of dideoxy ATP into the growing chain generates a series of smaller DNA fragments ending at all possible positions where adenine is found in the newly synthesized fragments. These correspond to positions where

thymine occurs in the original template strand. (B) An exposed X-ray film of a DNA sequencing gel. The four lanes represent A, C, G, and T dideoxy reaction mixes, respectively.

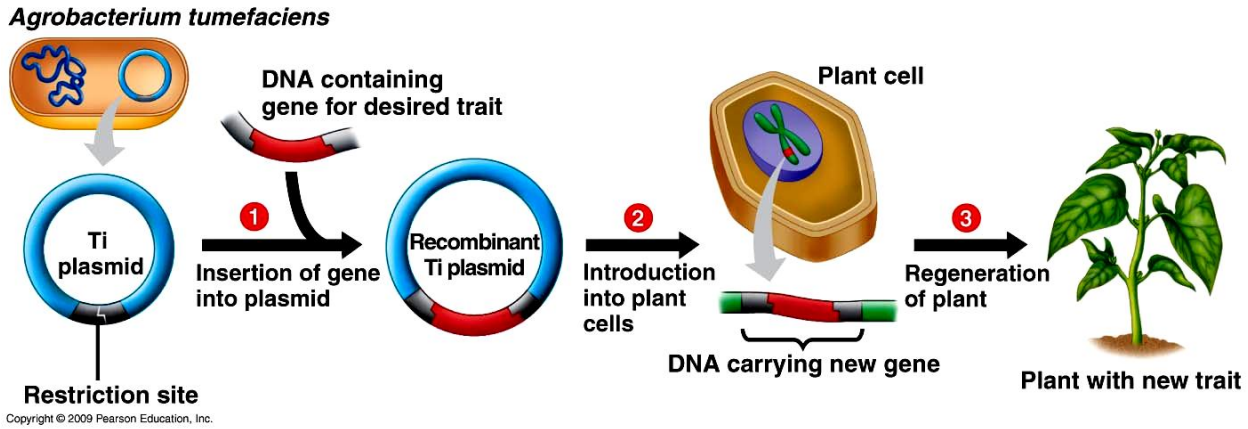
## AUTOMATED DNA SEQUENCING

In 1986, Leroy Hood and Lloyd Smith automated Sanger's method. In this new sequencing technology, radioactive markers are replaced with fluorescent ones. Each ddNTP terminator is tagged with a different color of fluorophore: red, green, blue, or yellow. Thus, instead of having to run four separate sequencing reactions, the reactions can be combined into one tube. The first automated sequencer made use of a polyacrylamide gel to resolve the samples, a laser to excite the dye molecules as they reached a detector near the end of the gel, and a computer to read the results as a DNA sequence. In this system each automated sequencer was able to produce 4800 bases of sequence per day. The current automated systems replace the old-style gel with arrays of tiny capilliaries, each of which acts as a "lane." A pump loads special capillaries with a polymer that serves as the separation matrix. DNA samples in a 96-well plate are loaded into the capillary array by a short burst of electrophoresis, called "electrokinetic injection." The capillary array is immersed in running buffer and the DNA fragments then migrate through the capillary matrix by size, smallest to largest. As the DNA fragments reach the detection window, a laser beam excites the dye molecules causing them to fluoresce. Emitted light from 96 capillaries is collected at once, spectrally separated into the four colors and focused onto a CCD camera. Computer software interprets the pattern of peaks to produce a graph of fluorescence intensity versus time (electropherogram), which is then converted to the DNA sequence. With this system, as many as 2 million bases can be sequenced per day.

## SOME IMPORTANT THINGS ABOUT RECOMBINANT DNA TECHNOLOGY

### $T_i$ PLASMID

**Ti Plasmid as Gene Vector** - Because of their natural ability to insert DNA into plant cells, the Ti plasmids have become the subject of extensive study for use in genetic engineering of plants. Ti plasmids are natural gene vectors which contain all the functions necessary for the transfer, stable incorporation and expression of genetic information in plants.



Studies on the signals involved and required for the transfer and stabilization of the T -DNA have revealed how the Ti plasmid can be used as a vector for insertion of foreign genes into the plant genome. First, the transposon Tn7 inserted in the T-region is co transferred to and maintained in the plant cell without major rearrangements.

The co-transfer of insertions at different positions in both the octopine and the nopaline T-region demonstrated involvement of "ends" for stabilization of the T-region. Therefore, one may expect that any segment between these "ends" would be co transferred and stabilized in the plant nucleus.

Moreover, studies of the functional organization of T-DNA have revealed that none of the genes involved in oncogenesis or suppression of differentiation are required for T -DNA transfer. However, the "ends", presumably including the DNA sequences around the 25 bp repeat box, have to occur in cis of the Ti plasmid.

A plasmid containing the T-region in trans of the Ti plasmid cannot induce tumors. Therefore, genes have to be engineered into the T-region of the Ti plasmids. Due to the large size of the Ti plasmids and the requirement of many genes for virulence, direct cloning into the Ti plasmid is impossible.

So a technique for shuttling genes into the Ti plasmid via an intermediate vector.was worked out. The intermediate vector contains a region of homology with the T-region of the Ti plasmid into which genes can be inserted. The engineered segments of the intermediate vector have been worked out.

One is derived from a broad host range vector (W- and P-type plasmids), capable of replicating stably in Agrobacterium. Recombinants between vector and Ti plasmid can be isolated by selecting for co-transfer of the cointegrate plasmid during conjugation.

This method requires conjugation in order to select for the first recombination, and subsequent screening for the second recombination or immediate selection by incompatibility for the substituted recombinants. The second type of intermediate vector is a pBR322 derivative, which can be mobilized efficiently by F- or I-type plasmids when the mob protein is complemented in trans (by a compatible ColE plasmid).

Since the pBR322-derivatives cannot replicate in Agrobacterium, they are maintained only through homologous cointegration with the Ti plasmid. The second recombinant (or substituted T-region) is obtained after propagation of cointegrates by screening for loss of the pBR322 vector markers. The three possibilities are outlined.

As the final goal in the breeding of normally engineered plants, gene transfer by the I-DNA has to be separated from its tumorigenic activities. This has been shown to be possible since none of the T-DNA genes are essential for T-region transfer. Regenerates are fertile and transmit T-DNA linked genes in a Mendelian way to the progeny.

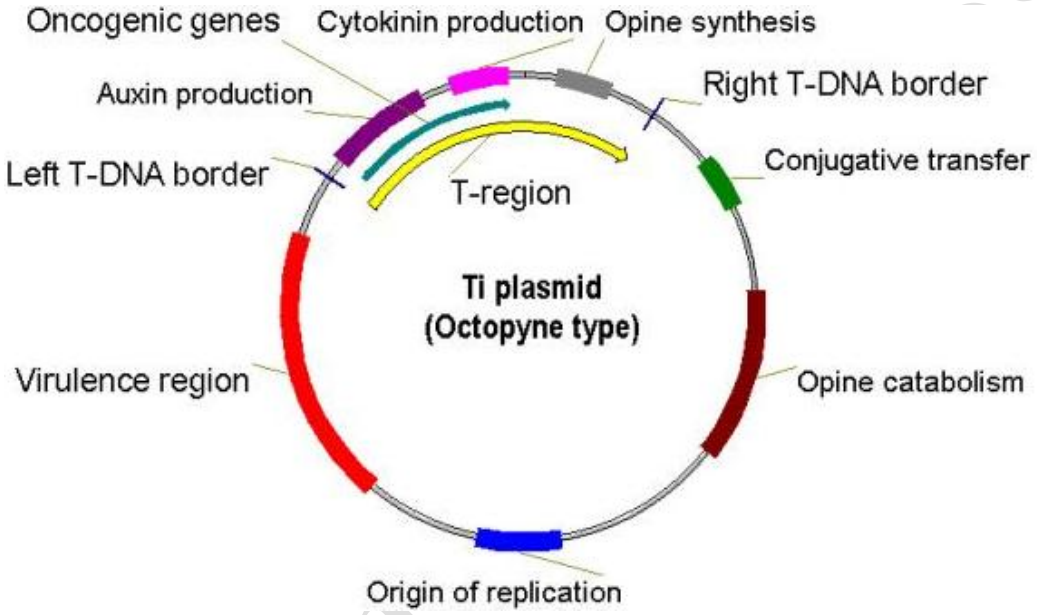
T-DNA can also be a source of expression signals for plant genes. Currently, T-DNA expression signals are being adapted for the insertion of foreign genes in an intermediate expression vector.

Career Avenues

Our knowledge about the Ti plasmids allows us to transfer any DNA sequence into plant cells, from which normal plants can be regenerated.

Recent work has shown that all that is needed for agrobacteria to infect and transform plant cells is an intact T-DNA region and another region of the Ti plasmid called 'vir'. More important from a practical standpoint is that these two regions do not have to be on the same plasmid.

Of an Agrobacterium harbors a Ti plasmid containing the vir region and another plasmid containing the T-DNA, the bacteria can transform plant cells and the T-DNA (and whatever other genes have been inserted into it) will be incorporated into the plant genome.




Career

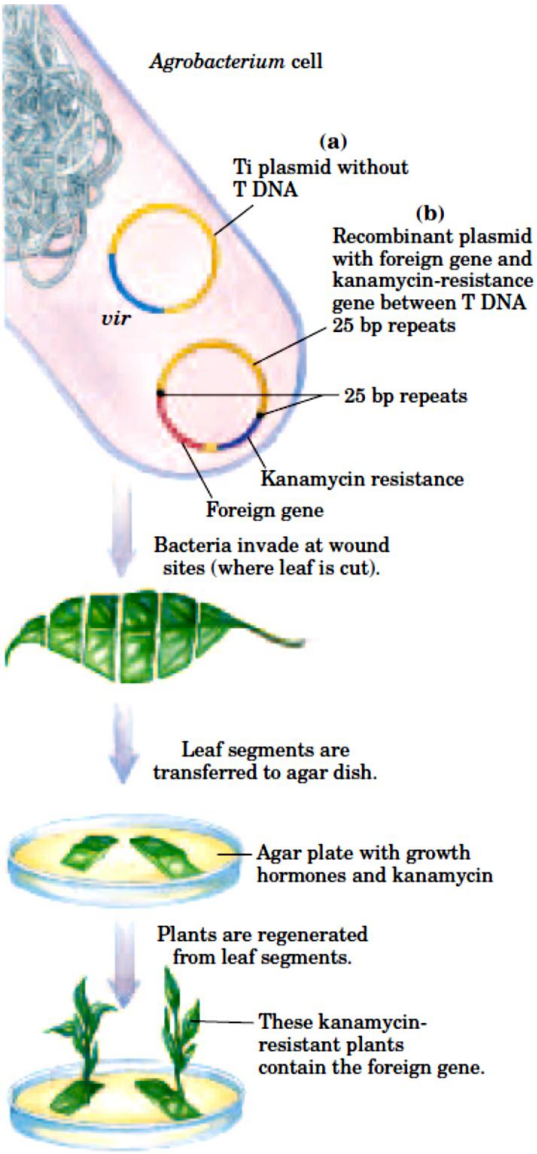



**Fig: A two-plasmid strategy to create a recombinant plant.**

**(a)** One plasmid is a modified Ti plasmid that contains the *vir* genes but lacks T DNA.

**(b)** The other plasmid contains a segment of DNA that bears both a foreign gene (the gene of interest) and an antibiotic resistance element (here, kanamycin resistance), flanked by the two 25 bp repeats of T DNA that are required for transfer of the plasmid genes to the plant chromosome. This plasmid also contains the replication origin needed for propagation in *Agrobacterium.* When bacteria invade at the site of a wound (the edge of the cut leaf ), the *vir* genes on the first plasmid mediate transfer into the plant genome of the segment of the second plasmid that is flanked by the 25 bp repeats. Leaf segments are placed on an agar dish that contains both kanamycin and appropriate levels of plant growth hormones, and new plants are generated from segments with the transformed cells. Nontransformed cells are killed by the kanamycin. The foreign gene and the antibiotic-resistance element are normally transferred together, so plant cells that grow in this medium generally contain the foreign gene.

# POLYMERASE CHAIN REACTION (PCR)

The polymerase chain reaction (PCR) is the one of the most powerful techniques that has been developed recently in the area of recombinant DNA research. PCR has had a major impact on many areas of molecular cloning and genetics. With this technique, a target sequence of DNA can be amplified a billion-fold in several hours. Amplification of particular segments of DNA by PCR is distinct from the amplification of DNA during cloning and propagation within a host cell. The procedure is carried out entirely *in vitro*. In addition to its use in many molecular cloning strategies, PCR is also used in the analysis of gene expression, forensic analysis where minute samples of DNA are isolated from a crime scene, and diagnostic tests for genetic diseases.

PCR is a DNA polymerase reaction. As with any DNA polymerase reaction it requires a DNA template and a free 3′-OH to get the polymerase started. The template is provided by the DNA sample to be amplified and the free 3′-OH groups are provided by site-specific oligonucleotides primers. The primers are complementary to each of the ends of the sequence that is to be amplified. Note that *in vivo* DNA polymerase would use an RNA primer, but a more stable, more easily synthesized DNA primer is used *in vitro*. The three steps of the reaction are denaturation, annealing of primers, and primer extension:

## DENATURATION

In the first step, the target sequence of DNA is heated to denature the template strands and render the DNA single-stranded.

## ANNEALING

The DNA is then cooled to allow the primers to anneal, that is, to bind the appropriate complementary strand. The temperature for this step varies depending on the size of the primer, the GC content, and its homology to the target DNA. Primers are generally DNA oligonucleotides of approximately 20 bases each.

## PRIMER EXTENSION

In the presence of Mg2+, DNA polymerase extends the primers on both strands from 5′ to 3′ by its polymerase activity. Primer extension is performed at a temperature optimal for the particular polymerase that is used. Currently, the most popular enzyme for this step is *Taq* polymerase, the DNA polymerase from the thermophilic (heat-loving) bacteria *Thermus aquaticus*. This organism lives in hot springs that can be near boiling and thus requires a thermostable polymerase.

These three steps are repeated from 28 to 35 times. With each cycle, more and more fragments are generated with just the region between the primers amplified. These accumulate exponentially. The contribution of strands with extension beyond the target sequence becomes negligible since these accumulate in a linear manner. After 25 cycles in an automated thermocycler machine, there is a $2^{25}$ amplification of the target sequence. PCR products can be

Career Avenues

visualized on a gel stained with nucleic acid-specific fluorescent compounds such as ethidium bromide or SYBR green. The error rate of *Taq* is 2 × 10−4. If an error occurs early on in the cycles, it could become prominent. Other polymerases, such as *Pfu*, have greater fidelity. *Pfu* DNA polymerase is from *Pyrococcus furiosus*. Base misinsertions that may occur infrequently during polymerization are rapidly excised by the 3′ → 5′ exonuclease (proofreading) activity of this enzyme.

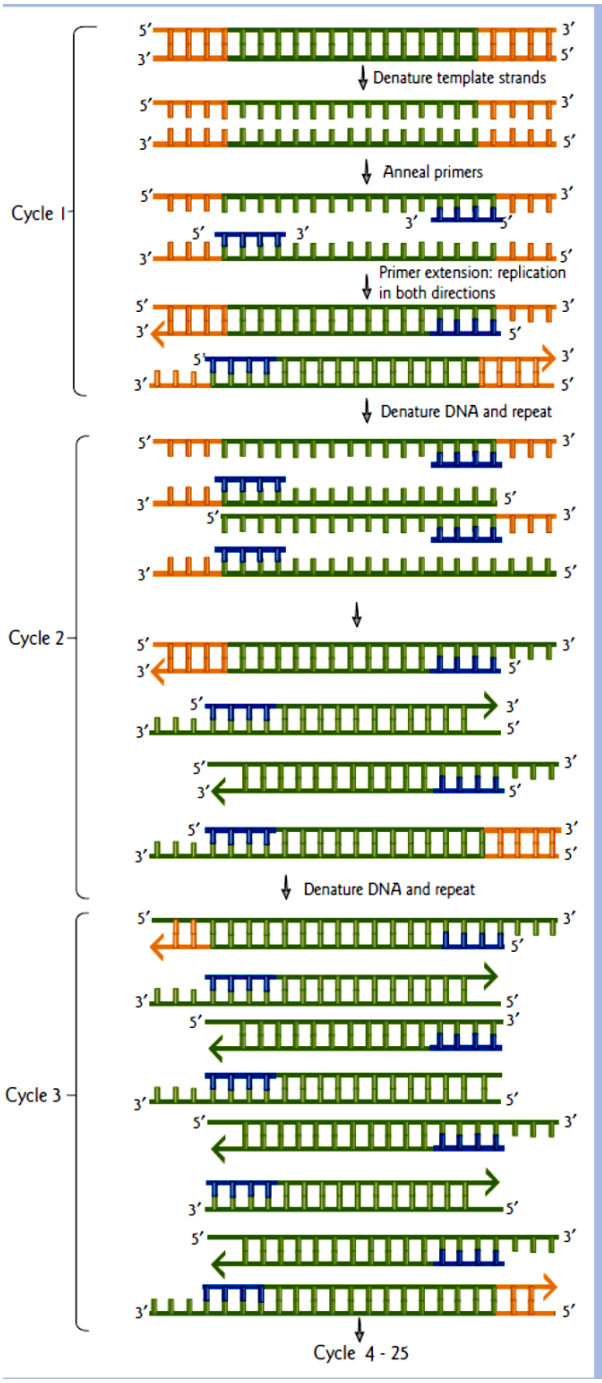



**Figure 1 Polymerase chain reaction (PCR).** PCR is an *in vitro* DNA replication method. The starting material is a double-stranded DNA. The target sequence to be amplified is indicated in green. Large numbers of two primers (blue) are added, each with a sequence complementary to that found in one strand at the end of the region to be amplified. A thermostable DNA polymerase (e.g. *Taq* polymerase) and dNTPs are also added. In the first cycle, heating to 95°C denatures the double-stranded DNA and subsequent cooling to 55–65°C then allows the primers to anneal to their complementary sequences in the target DNA. *Taq* polymerase extends each primer from 5′ to 3′, generating newly synthesized strands in both directions, which extend to the end of the template strands. The extension is performed at 72°C. In the second cycle, the original and newly made DNA strands are denatured at 95°C and primers are annealed to their complementary sequences at 55–65°C. Each annealed primer again is extended by *Taq* polymerase. In the third cycle, two double-strand DNA molecules are generated exactly equal to the target sequence. These two are doubled in the fourth cycle and are doubled again with each successive cycle.

## RT-PCR

The thermostable polymerase used in the basic PCR requires a DNA template and hence is limited to the amplification of DNA samples. There are numerous instances in which the amplification of RNA would be preferred. For example, in analyses involving the diffierential expression of genes in tissues during development or the cloning of DNA derived from an mRNA (complementary DNA or *cDNA*), particularly a rare mRNA. In order to apply PCR methodology to the study of RNA, the RNA sample must first be reverse-transcribed to cDNA to provide the necessary DNA template for the thermostable polymerase. This process is called reverse transcription (RT), hence the name RT-PCR. Avian myeloblastosis virus (AMV) or Moloney murine leukaemia virus (MuLV) reverse transcriptases are generally used to produce a DNA copy of the RNA template. Various strategies can be adopted for first-strand cDNA synthesis.

## LONG ACCURATE PCR (LA-PCR)

Amplification of long DNA fragments is desirable for numerous applications of gene manipulation. The basic PCR works well when small fragments are amplified. The efficiency of amplification and therefore the yield of amplified fragments decrease significantly as the size of the amplicon increases over 5 kb.

This decrease in yield of longer amplified fragments is attributable to partial synthesis across the desired sequence, which is not a suitable substrate for the subsequent cycles. This is demonstrated by the presence of smeared, as opposed to discrete, bands on a gel. Some scientists examined the factors affecting the thermostable polymerization across larger regions of DNA and identified key variables affecting the yield of longer PCR fragments.

Most significant of these was the absence of a 3'-5'exonuclease (proofreading) activity in *Taq*polymerase. Presumably, when the *Taq*polymerase misincorporates a dNTP, subsequent extension of the strand either proceeds very slowly or stops completely. To overcome this problem, a second thermostable polymerase with proofreading capability is added.

## KEY FACTORS AFFECTING THE PCR

The specificity of the PCR depends crucially upon the primers. The following factors are important in choosing effective primers:

- Primers should be 17 to 30 nucleotides in length.
- A GC content of about 50% is ideal. For primers with a low GC content, it is desirable to choose a long primer so as to avoid a low melting temperature.
- Sequences with long runs (i.e. more than three or four) of a single nucleotide should be avoided.
- Primers with significant secondary structure are undesirable.

Career Avenues

- There should be no complementarity between the two primers. The great majority of primers which conform to these guidelines can be made to work, although not all comparable primer sets are equally effective even under optimized conditions.

## REAL-TIME QUANTITATIVE PCR

Instrumentation has been developed which combines thermal cycling with measurement of fluorescence, thereby enabling the progress of the PCR to be monitored in real time. This revolutionizes the way one approaches PCR-based quantitation of DNA. Reactions are characterized by the point in time during cycling when amplification of a product is first detected, rather than by the amount of PCR product accumulated after a fixed number of cycles. The higher the starting copy number of the target, the sooner a significant increase in fluorescence is noted. Quantitation of the amount of target in unknown samples is achieved by preparing a standard curve, using different starting copy numbers of the target sequence.

Career Avenues

# ELECTROPHORESIS

Electrophoresis is the standard method for analyzing, identifying, and purifying fragments of DNA or RNA that differ in size, charge, or conformation. It is one of the most widely used techniques in molecular biology. Walk into any molecular biology lab, and the odds are you will see at least one gel apparatus in operation. When charged molecules are placed in an electric field, they migrate toward the positive (anode, red) or negative (cathode, black) pole according to their charge. In contrast to proteins, which can have either a net positive or net negative charge, nucleicacids have a consistent negative change due to their phosphate backbone, and they migrate toward the anode.

Proteins and nucleic acids are separated by electrophoresis within a matrix or "gel." Most commonly, the gel is cast in the shape of a thin slab, with wells for loading the sample. The gel is immersed within an electrophoresis buffer that provides ions to carry a current and some type of buffer to maintain the pH at a relatively constant value. The gels used for eletrophoresis are composed either of agarose or polyacrylamide. Agarose gels are used in a horizontal gel apparatus, while polyacrylamide gels are used in a vertical gel apparatus. These two differ in resolving power. Agarose gels are used for the analysis and preparation of fragments between 100 and 50,000 bp in size with moderate resolution, and polyacrylamide gels are used for the analysis and preparation of small molecules with single nucleotide resolution.

This high resolution is required for applications such as for DNA sequencing, DNase I footprinting, and electrophoretic mobility shift assays. In contrast to agarose, polyacrylamide gels are also widely used for the electrophoresis of proteins.

## AGAROSE GEL ELECTROPHORESIS

Agarose is a polysaccharide extracted from seaweed. Agarose gels are easily prepared by mixing agarose powder with buffer solution, boiling in a microwave to melt, and pouring the gel into a mold where the agarose (generally 0.5–2.0%) solidifies into a slab. Agarose gels have a large range of separation, but relatively low resolving power. By varying the concentration of agarose, fragments of DNA from about 100 to 50,000 bp can be separated using standard electrophoretic techniques. A toothed comb forms wells in the agarose. The agarose slab is submerged in a buffer solution and an electric current is passed through the gel, with the negatively charged DNA (due to the phosphate in the sugar–phosphate backbone) moving through the gel from the negatively charged electrode (cathode) towards the positive electrode (anode). Pores between the agarose molecules act like a sieve that separates the molecules by size. In gel filtration chromatography, nucleic acids flow around the spherical agarose beads. In contrast, in an electrophoretic gel, nucleic acids migrate through the pores; thus fragments separate by size with the smallest pieces moving the fastest and farthest through the gel. Because DNA by itself is not visible in the gel, the DNA is stained with a fluorescent dye such as ethidium bromide (EtBr). EtBr intercalates between the bases causing DNA to fluoresce orange when the dye is illuminated by ultraviolet light.

Career Avenues

## PULSED FIELD GEL ELECTROPHORESIS (PFGE)

DNA changes conformation as it moves through gels, alternating between extended and compact forms. The poured between glass plates. Gels can be either nondenaturing or denaturing (e.g. contain 8 M urea). Denaturing gels are used, for example, when single stranded DNA is being analyzed, as in DNA sequencing. Polyacrylamide gels have a rather small range of separation, but very high resolving power. In the case of DNA, polyacrylamide is used for separating fragments of less than about 500 bp. However, under appropriate conditions, fragments of DNA differing in length by a single base pair are easily resolved. Bands in polyacrylamide gels are usually detected by autoradiography, although silver staining can also be used.

Mobility of a DNA molecule depends upon the relationship between the pore size of the gel and the globular size of the DNA in its compact form, with larger molecules moving more slowly. Once DNA reaches a critical size, the molecule is too large to fit through any of the pores in an agarose gel and can move only as an extended molecule. The mobility of the DNA thus becomes independent of size, resulting in comigration of all large molecules. To fractionate large DNA molecules such as YACs, agarose gel electrophoresis is carried out with a pulsed electric field. The periodic field causes the DNA molecule to reorient; longer molecules take longer to realign than shorter ones, thus delaying their progress through the gel and allowing them to be resolved. DNA molecules up to 200–400 Mb in size have been separated by various pulsed field-based methods.

## POLYACRYLAMIDE GEL ELECTROPHORESIS (PAGE)

Polyacrylamide is a cross-linked polymer of acrylamide. The length of the polymer chains is dictated by the concentration of acrylamide used, which is typically between 3.5 and 20%. Polyacrylamide gels are significantly more cumbersome to prepare than agarose gels. Because oxygen inhibits the polymerization process, they must be poured between glass plates. Gels can be either nondenaturing or denaturing (e.g. contain 8 M urea). Denaturing gels are used, for example, when singlestranded DNA is being analyzed, as in DNA sequencing. Polyacrylamide gels have a rather small range of separation, but very high resolving power. In the case of DNA, polyacrylamide is used for separating fragments of less than about 500 bp. However, under appropriate conditions, fragments of DNA differing in length by a single base pair are easily resolved. Bands in polyacrylamide gels are usually detected by autoradiography, although silver staining can also be used.

Career Avenues

# DNA FINGERPRINTING

## WHAT IS DNA FINGERPRINTING?

It is a technique, by which an individual can be identified at molecular level. With the advancement of science and technology STR analysis has become very popular in forensic laboratories. Scientists have chosen repeating sequences in the DNA, which are present in all individuals on different chromosomes, and are known to vary from individual to individual except in identical twins. These are used as genetic markers to identify the individual.

## WHAT IS THE USE OF DNA FINGERPRINTING TECHNOLOGY?

DNA fingerprinting technology has made it possible to identify the source of biological samples found at scenes of crime. This will resolve disputes of maternity /paternity, identification of mutilated remains, identification of rape/ murder, identification of missing child, exchange of babies in hospital wards, forensic wildlife, protection of farmers' rights and biodiversity. This remarkable technology provides exclusion as well as positive identification with virtually 100% precision.

## MAKING DNA FINGERPRINTS

DNA fingerprinting is a laboratory procedure that requires six steps:

- **Isolation of DNA:** DNA must be recovered from the cells or tissues of the body. Only a small amount of tissue - like blood, hair, or skin - is needed. For example, the amount of DNA found at the root of one hair is usually sufficient.
- **Cutting, sizing, and sorting:** Special enzymes called restriction enzymes are used to cut the DNA at specific places. For example, an enzyme called EcoR1, found in bacteria, will cut DNA only when the sequence GAATTC occurs. The DNA pieces are sorted according to size by a sieving technique called electrophoresis. The DNA pieces are passed through a gel made from seaweed agarose (a jelly-like product made from seaweed). This technique is the biotechnology equivalent of screening sand through progressively finer mesh screens to determine particle sizes.
- **Transfer of DNA to nylon:** The distribution of DNA pieces is transferred to a nylon sheet by placing the sheet on the gel and soaking them overnight.
- **Probing**: Adding radioactive or colored probes to the nylon sheet produces a pattern called the DNA fingerprint. Each probe typically sticks in only one or two specific places on the nylon sheet.

Career Avenues

- **DNA fingerprint:** The final DNA fingerprint is built by using several probes (5-10 or more) simultaneously. It resembles the bar codes used by grocery store scanners.

Career Avenues

# SOUTHERN BLOTTING AND NORTHERN BLOTTING

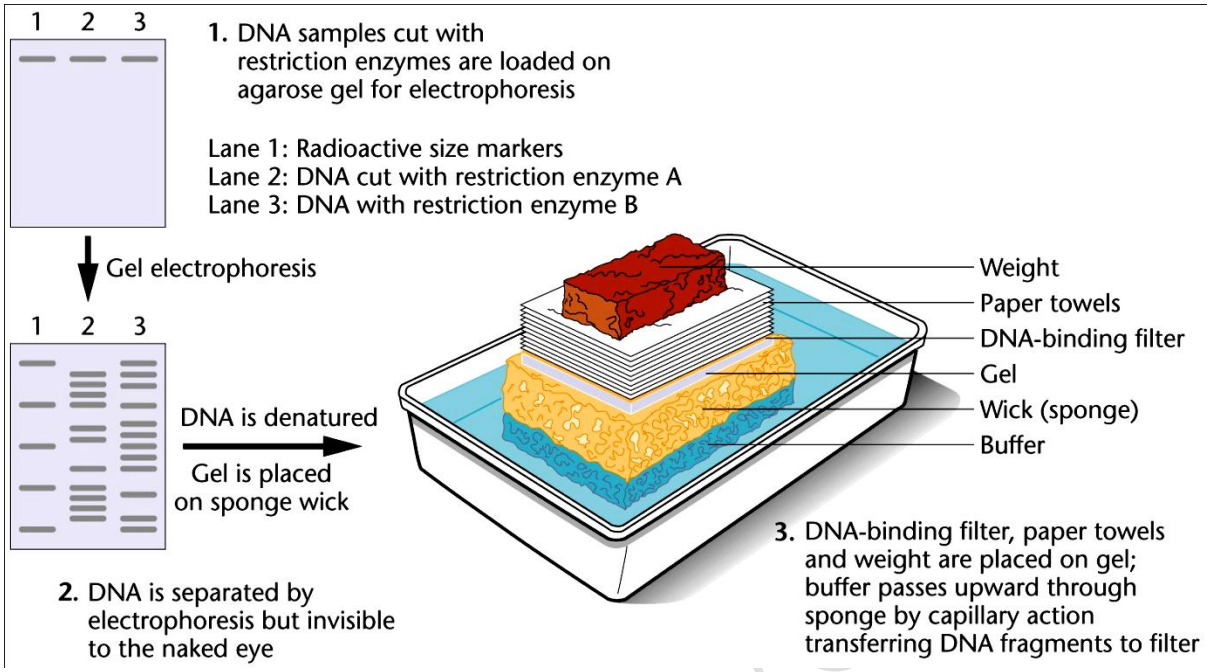
In the procedure, referred to as **Southern blotting**, the DNA fragments separated on an agarose gel are transferred and immobilized onto a membrane. After the gel has been run, the DNA fragments are denatured (i.e. the strands are separated) using alkali. The single-stranded nature of the DNA on the membrane is important to allow complementary DNA sequences to be able to bind to the DNA fragments attached to the membrane. After the DNA in the gel has been denatured, the DNA fragments are transferred to a membrane. The transfer process can be mediated either electrophoretically or through capillary attraction by placing the gel and membrane in contact with each other and allowing buffer to flow through the gel onto the membrane – the DNA fragments move with the buffer and become trapped on the membrane.

Initially, nitrocellulose membranes were used, but these were fragile and easily broken. Nylon membranes are commonly used today. After transfer, the DNA fragments need to be fixed to the membrane so that they cannot detach. A number of methods of fixing are available including baking at 80 ◦C and ultraviolet cross-linking. UV cross-linking is based on the formation of crosslinks between a small fraction of the T residues in the DNA and the positively charged and amino groups on the surface of the nylon membrane. Following fixation, the membrane is placed in a solution of labelled (often radioactive) single-stranded nucleic acid – either single-stranded DNA or RNA. The labelled nucleic acid (or **probe**) is allowed to hybridize to its complementary partner sequence on the membrane. The interaction between the single-stranded probe and its complementary sequence will result in the binding of the probe to the membrane through non-covalent hydrogen bonding that normally holds the DNA double helix together. The membrane is then washed extensively to remove non-specifically bound probe, and specific interactions are detected by exposing the membrane to X-ray film.

Southern blotting is used to detect DNA sequences that are either identical or similar to the sequence of the probe. The hybridization of the probe to the DNA sequences trapped on the membrane is the critical component of success of these experiments. As we have seen previously, single-stranded DNA will bind with high affinity to its complementary partner sequence. It will also bind to similar, but non-identical, DNA sequences with a reduced affinity. This differential binding affinity to different DNA sequences can be used to identify DNA molecules that are not identical but are merely related to the probe. This sort of analysis is achieved by altering the temperature (or salt concentration) at which the probe is washed off the filters after it has bound. Washing the membrane at high temperature (high stringency) will result in thermal disruption of all but the most tightly bound sequences. Consequently, the bands that show up on the X-ray will film will be either identical or highly related to the probe. Washing the membrane to lower temperatures will reduced the overall level of stringency, and sequences that are less related to the probe will give positive signals on the X-ray film.

Career Avenues

## SOUTHERN BLOT ANALYSIS




## NORTHERN BLOTTING

The detection of RNA sequences using probes. RNA is separated on an agarose gel and transferred to a membrane, much as described above. Specific RNA molecules are then detected by hybridization using labelled single-stranded DNA or RNA sequences that are complementary to particular RNAs.

## WESTERN BLOTTING

The detection of specific proteins using antibodies. Proteins are separated through a polyacrylamide gel containing the detergent SDS to keep them in an unfolded (**denatured**) state. The proteins are transferred from the gel onto a membrane in much the same way as described above for Southern blotting. Particular proteins are then detected using antibodies. The specific interaction between the antibody and its antigen occurs on the membrane, and the position of the bound antibody is detected. Initially radio-labelled antibodies were used, but these have been largely superseded by antibody 'sandwiches'. The sandwiches work through the binding of one unlabelled antibody (the primary antibody) to the antigen on the membrane. A second, labelled, antibody (the secondary antibody) is then used to detect the presence of the first antibody. This has several advantages; firstly, the multivalent nature of antibody binding means that a substantial increase in sensitivity is achieved, and secondly, a single secondary antibody can be used to detect a number of different primary antibodies. For example, primary monoclonal antibodies are often raised in mice. A secondary antibody raised, say, in the rabbit against mouse immunoglobulins will be capable of interacting with a number of different mouse derived primary antibodies.

Career Avenues

## SOUTH-WESTERN BLOTTING

The identification of proteins that bind to particular DNA sequences. Proteins are either separated on a gel or blotted directly onto membranes. The membrane is then challenged with a labelled double-stranded DNA oligonucleotide. If a protein is able to bind the DNA, it will be detected by the presence of the label.

Career Avenues

# *IN-SITU* HYBRIDIZATION

In situ hybridization (ISH) is a type of hybridization that uses a labeled complementary DNA or RNA strand (i.e., probe) to localize a specific DNA or RNA sequence in a portion or section of tissue (in situ), or, if the tissue is small enough (e.g. plant seeds, Drosophila embryos), in the entire tissue (whole mount ISH). This is distinct from immunohistochemistry, which usually localizes proteins in tissue sections.

## *IN SITU* HYBRIDIZATION PROBES

- double-stranded DNA (dsDNA) probes
- single-stranded DNA (ssDNA) probes
- RNA probes (riboprobes)
- synthetic oligonucleotides (PNA, LNA)

## TYPES OF IN-SITU HYBRIDIZATION

- **DNA ISH** can be used to determine the structure of chromosomes.
- **Fluorescent DNA ISH (FISH)** can, for example, be used in medical diagnostics to assess chromosomal integrity.
- **RNA ISH** (hybridization histochemistry) is used to measure and localize mRNAs and other transcripts within tissue sections or whole mounts.

## APPLICATIONS OF IN SITU HYBRIDIZATION

- **Microbiology** (classic target - 16S rRNA) - morphology and population structure of microorganisms
- **Pathology** (pathogen profiling, abnormal gene expression)
- **Developmental biology** (gene expression profiling in embryonic tissues)
- **Karyotyping and phylogenetic analysis** (unique FISH patterns on individual chromosomes, chromosomal abberations)
- **Physical mapping** (mapping clones on chromosomes and direct assignment of mapped clones to chromosomal regions associated with heterochromatin or euchromatin)

Career Avenues

# FROM GENOMES TO PROTEOMES

A gene is not simply a DNA sequence; it is information that is converted to a useful product—a protein or functional RNA molecule—when and if needed by the cell. The first and most obvious step in exploring a large sequenced genome is to catalog the products of the genes within that genome. Genes that encode RNA as their final product are somewhat harder to identify than are protein-encoding genes, and even the latter can be very difficult to spot in a vertebrate genome. The explosion of DNA sequence information has also revealed a sobering truth. Despite many years of biochemical advances, there are still thousands of proteins in every eukaryotic cell (and quite a few in bacteria) that we know nothing about. These proteins may have functions in processes not yet discovered, or may contribute in unexpected ways to processes we think we understand. In addition, the genomic sequences tell us nothing about the three dimensional structure of proteins or how proteins are modified after they are synthesized. The proteins, with their myriad critical functions in every cell, are now becoming the focus of new strategies for whole cell biochemistry.

The complement of proteins expressed by a genome is called its **proteome,** a term that first appeared in the research literature in 1995. This concept rapidly evolved into a separate field of investigation, called **proteomics.** The problem addressed by proteomics research is straightforward, although the solution is not. Each genome presents us with thousands of genes encoding proteins, and ideally we want to know the structure and function of all those proteins. Given that many proteins offer surprises even after years of study, the investigation of an entire proteome is a daunting enterprise. Simply discovering the function of new proteins requires intensive work. Biochemists can now apply shortcuts in the form of a broad array of new and updated technologies. Protein function can be described on three levels.

- **Phenotypic function** describes the effects of a protein on the entire organism. For example, the loss of the protein may lead to slower growth of the organism, an altered development pattern, or even death.
- **Cellular function** is a description of the network of interactions engaged in by a protein at the cellular level. Interactions with other proteins in the cell can help define the kinds of metabolic processes in which the protein participates.
- **Molecular function** refers to the precise biochemical activity of a protein, including details such as the reactions an enzyme catalyzes or the ligands a receptor binds.

For several genomes, such as those of the yeast *Saccharomyces cerevisiae* and the plant *Arabidopsis*, a massive effort is underway to inactivate each gene by genetic engineering and to investigate the effect on the organism. If the growth patterns or other properties of the organism change (or if it does not grow at all), this provides information on the phenotypic function of the protein product of the gene.

There are three other main paths to investigating protein function:

Career Avenues

- Sequence and structural comparisons with genes and proteins of known function,
- Determination of when and where a gene is expressed, and
- Investigation of the interactions of the protein with other proteins.

## SEQUENCE OR STRUCTURAL RELATIONSHIPS PROVIDE INFORMATION ON PROTEIN FUNCTION

One of the important reasons to sequence many genomes is to provide a database that can be used to assign gene functions by genome comparisons, an enterprise referred to as **comparative genomics.** Sometimes a newly discovered gene is related by sequence homologies to a gene previously studied in another or the same species, and its function can be entirely or partly defined by that relationship. Such genes—of different species but possessing a clear sequence and functional relationship to each other — are called **orthologs.** Genes similarly related to each other within a single species are called **paralogs**. If the function of a gene has been characterized for one species, this information can be used to assign gene function to the ortholog found in the second species. The identity is easiest to make when comparing genomes from relatively closely related species, such as mouse and human, although many clearly orthologous genes have been identified in species as distant as bacteria and humans. Sometimes even the order of genes on a chromosome is conserved over large segments of the genomes of closely related species. Conserved gene order, called **synteny,** provides additional evidence for an orthologous relationship between genes at identical locations within the related segments.

## CELLULAR EXPRESSION PATTERNS CAN REVEAL THE CELLULAR FUNCTION OF A GENE

In every newly sequenced genome, researchers find genes that encode proteins with no evident structural relationships to known genes or proteins. In these cases, other approaches must be used to generate information about gene function. Determining which tissues a gene is expressed in, or what circumstances trigger the appearance of the gene product, can provide valuable clues. Many different approaches have been developed to study these patterns.

## TWO-DIMENSIONAL GEL ELECTROPHORESIS

two-dimensional gel electrophoresis allows the separation and display of up to 1,000 different proteins on a single gel. Mass spectrometry can then be used to partially sequence individual protein spots and assign each to a gene. The appearance and nonappearance (or disappearance) of particular protein spots in samples from different tissues, from similar tissues at different stages of development, or from tissues treated in ways that simulate a variety of biological conditions can help define cellular function.

Career Avenues

## DNA MICROARRAYS

Major refinements of the technology underlying DNA libraries, PCR, and hybridization have come together in the development of **DNA microarrays** (sometimes called **DNA chips**), which allow the rapid and simultaneous screening of many thousands of genes. DNA segments from known genes, a few dozen to hundreds of nucleotides long, are amplified by PCR and placed on a solid surface, using robotic devices that accurately deposit nanoliter quantities of DNA solution. Many thousands of such spots are deposited in a predesigned array on a surface area of just a few square centimeters. An alternative strategy is to synthesize DNA directly on the solid surface, using photolithography. Once the chip is constructed, it can be probed with mRNAs or cDNAs from a particular cell type or cell culture to identify the genes being expressed in those cells.

A microarray can answer such questions as which genes are expressed at a given stage in the development of an organism. The total complement of mRNA is isolated from cells at two different stages of development and converted to cDNA, using reverse transcriptase and fluorescently labeled deoxynucleotides. The fluorescent cDNAs are then mixed and used as probes, each hybridizing to complementary sequences on the microarray. In given figure, for example, the labeled nucleotides used to make the cDNA for each sample fluoresce in two different colors. The cDNA from the two samples is mixed and used to probe the microarray. Spots that fluoresce green represent mRNAs more abundant at the single-cell stage; those that fluoresce red represent sequences more abundant later in development. The mRNAs that are equally abundant at both stages of development fluoresce yellow. By using a mixture of two samples to measure relative rather than absolute abundance of sequences, the method corrects for variations in the amounts of DNA originally deposited in each spot on the grid and other possible inconsistencies among spots in the microarray. The spots that fluoresce provide a snapshot of all the genes being expressed in the cells at the moment they were harvested—gene expression examined on a genome-wide scale. For a gene of unknown function, the time and circumstances of its expression can provide important clues about its role in the cell.

Career Avenues

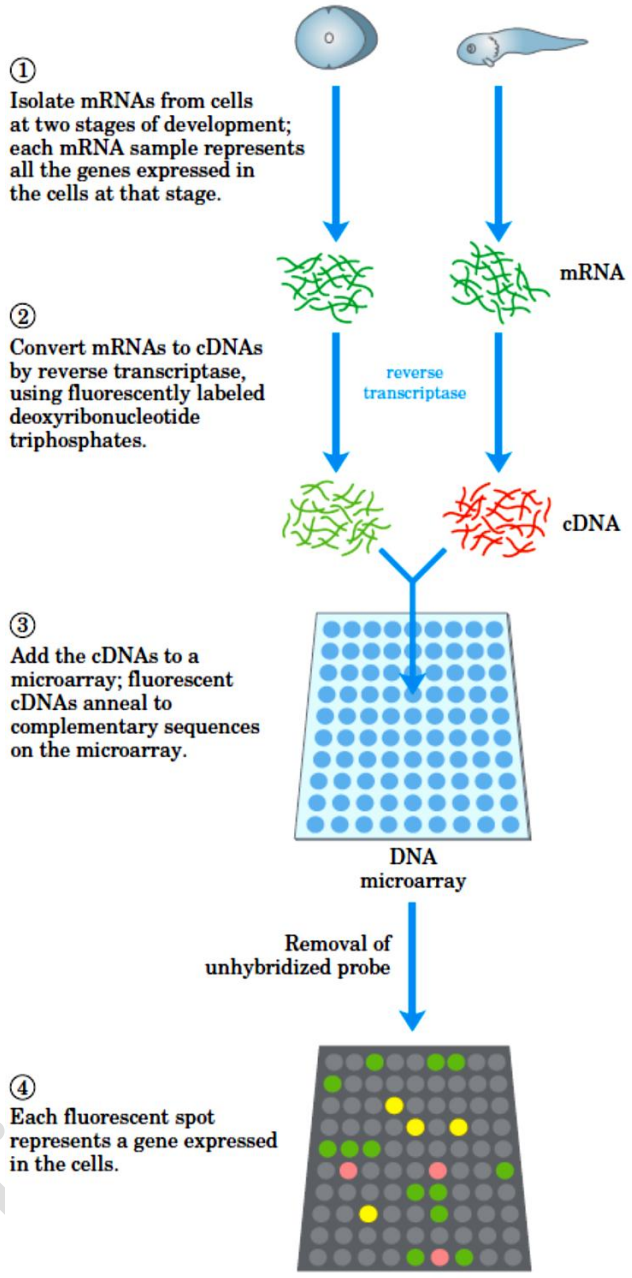



**Figure: DNA microarray.** A microarray can be prepared from any known DNA sequence, from any source, generated by chemical synthesis or by PCR. The DNA is positioned on a solid surface (usually specially treated glass slides) with the aid of a robotic device capable of depositing very small (nanoliter) drops in precise patterns. UV light cross-links the DNA to the glass slides. Once the DNA is attached to the surface, the microarray can be probed with other fluorescently labeled nucleic acids. Here, mRNA samples are collected from cells at two different stages in the development of a frog. The cDNA probes are made with nucleotides that fluoresce in different colors for each sample; a mixture of the cDNAs is used to probe the microarray. Green spots represent mRNAs more abundant at the single-cell stage; red

spots, sequences more abundant later in development. The yellow spots indicate approximately equal abundance at both stages.

## PROTEIN CHIPS

Proteins, too, can be immobilized on a solid surface and used to help define the presence or absence of other proteins in a sample. For example, researchers prepare an array of antibodies to particular proteins by immobilizing them as individual spots on a solid surface. A sample of proteins is added, and if the protein that binds any of the antibodies is present in the sample, it can be detected by a solid-state form of the ELISA assay. Many other types and applications of protein chips are being developed.

## YEAST TWO-HYBRID ANALYSIS TO DEFINING PROTEIN-PROTEIN INTERACTIONS

A sophisticated genetic approach to defining protein-protein interactions is based on the properties of the Gal4 protein (Gal4p), which activates transcription of certain genes in yeast. Gal4p has two domains, one that binds to a specific DNA sequence and another that activates the RNA polymerase that synthesizes mRNA from an adjacent reporter gene. The domains are stable when separated, but activation of the RNA polymerase requires interaction with the activation domain, which in turn requires positioning by the DNA-binding domain. Hence, the domains must be brought together to function correctly.

In this method, the protein-coding regions of genes to be analyzed are fused to the coding sequences of either the DNA-binding domain or the activation domain of Gal4p, and the resulting genes express a series of fusion proteins. If a protein fused to the DNA-binding domain interacts with a protein fused to the activation domain, transcription is activated. The reporter gene transcribed by this activation is generally one that yields a protein required for growth, or is an enzyme that catalyzes a reaction with a colored product. Thus, when grown on the proper medium, cells that contain such a pair of interacting proteins are easily distinguished from those that do not. Typically, many genes are fused to the Gal4p DNA-binding domain gene in one yeast strain, and many other genes are fused to the Gal4p activation domain gene in another yeast strain, then the yeast strains are mated and individual diploid cells grown into colonies. This allows for large-scale screening for proteins that interact in the cell.

Career Avenues

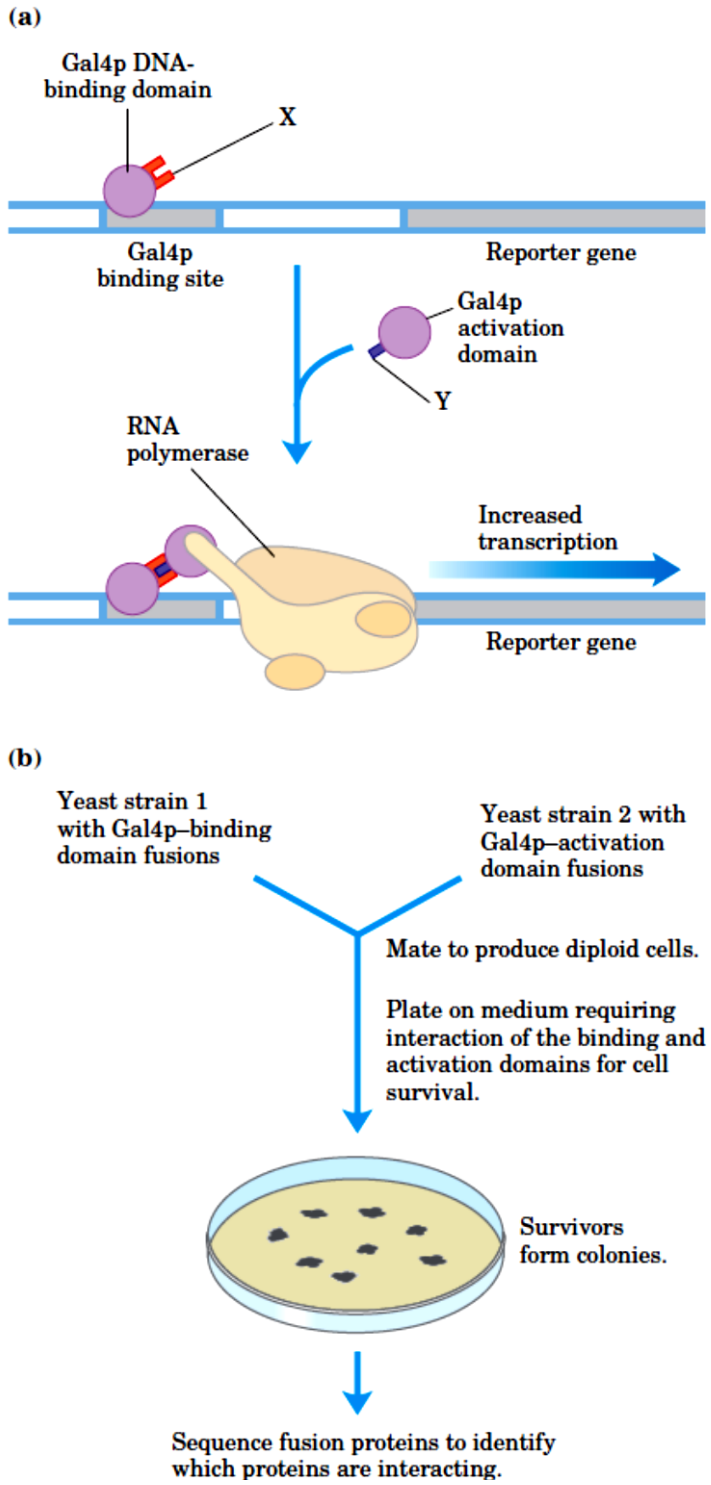



**Figure: The yeast two-hybrid system. (a)** In this system for detecting protein-protein interactions, the aim is to bring together the DNA-binding domain and the activation domain of the yeast Gal4 protein through the interaction of two proteins, X and Y, to which each domain is fused. This interaction is accompanied by the expression of a reporter gene. **(b)** The two fusions are created in separate yeast strains, which are then mated. The mated mixture is plated on a medium on which the yeast cannot

survive unless the reporter gene is expressed. Thus, all surviving colonies have interacting protein fusion pairs. Sequencing of the fusion proteins in the survivors reveals which proteins are interacting.

Career Avenues

# SITE-DIRECTED MUTAGENESIS

## CHANGING GENES: SITE-DIRECTED MUTAGENESIS

Three different methods of site-directed mutagenesis have been devised: cassette mutagenesis, primer extension and procedures based on the PCR. In some cases, the goal of protein engineering is to generate a molecule with an improvement in some operating parameter, but it is not known what amino acid changes to make. In this situation, a random mutagenesis strategy provides a route to the desired protein. However, methods based on gene manipulation differ from traditional mutagenesis in that the mutations are restricted to the gene of interest or a small portion of it. Genetic engineering also provides a number of simple methods of generating chimeric proteins where each domain is derived from a different protein. It should not be forgotten that constructing the mutant DNA is only part of the task. The vector for expression, the expression system and strategies for purification and assay must also be considered before embarking on protein mutagenesis.

## CASSETTE MUTAGENESIS

In cassette mutagenesis, a synthetic DNA fragment containing the desired mutant sequence is used to replace the corresponding sequence in the wild-type gene. This method was originally used to generate improved variants of the enzyme subtilisin. It is a simple method for which the efficiency of mutagenesis is close to 100%. The disadvantages are the requirement for unique restriction sites flanking the region of interest and the limitation on the realistic number of different oligonucleotide replacements that can be synthesized. The latter problem can be minimized by the use of doped oligonucleotides.

## PRIMER EXTENSION: THE SINGLE-PRIMER METHOD

The simplest method of site-directed mutagenesis is the single-primer method. The method involves priming *in vitro* DNA synthesis with a chemically synthesized oligonucleotide (7–20 nucleotides long) that carries a base mismatch with the complementary sequence. The method requires that the DNA to be mutated is available in single-stranded form, and cloning the gene in M13-based vectors makes this easy. However, DNA cloned in a plasmid and obtained in duplex form can also be converted to a partially single-stranded molecule that is suitable The synthetic oligonucleotide primes DNA synthesis and is itself incorporated into the resulting heteroduplex molecule. After transformation of the host *E. coli*, this heteroduplex gives rise to homoduplexes whose sequences are either that of the original wild-type DNA or that containing the mutated base.

The frequency with which mutated clones arise, compared with wild-type clones, may be low. In order to pick out mutants, the clones can be screened by nucleic acid hybridization with $^{32}$P-

Career Avenues

labelled oligonucleotide as probe. Under suitable conditions of stringency, i.e. temperature and cation concentration, a positive signal will be obtained only with mutant clones. This allows ready detection of the desired mutant. In order to check that the procedure has not introduced other adventitious changes, it is prudent to check the sequence of the mutant directly by DNA sequencing. This was a particular necessity with early versions of the technique which made use of *E. coli* DNA polymerase. The more recent use of the high-fidelity DNA polymerases from phages T4 and T7 has minimized the problem of extraneous mutations, as well as shortening the time for copying the second strand. Also, these polymerases do not 'strand-displace' the oligomer, a process which would eliminate the original mutant oligonucleotide.

## PCR METHODS OF SITE-DIRECTED MUTAGENESIS

The limitation of restriction sites in cassette mutagenesis may be overcome using polymerase chain reaction with oligonucleotide "primers", such that a larger fragment may be generated covering two convenient restriction sites. The exponential amplification in PCR produces a fragment containing the desired mutation in sufficient quantity to be separate from the original, unmutated plasmid by gel electrophoresis, which may then be inserted in the original context using standard recombinant molecular biology techniques. There are many variations of the same technique. The simplest method places the mutation site towards one of the ends of the fragment whereby one of two oligonucleotides used for generating the fragment contains the mutation. This involves a single step of PCR, but still has the inherent problem of requiring a suitable restriction site near the mutation site unless a very long primer is used. Other variations therefore employ three or four oligionucleotides, two of which may be non-mutagenic oligonucleotides that cover two convenient restriction sites and generate a fragment that can be digested and ligated into a plasmid, while the mutagenic oligonucleotide may be complementary to a location within that fragment well away from any convenient restriction site. These methods require multiple steps of PCR so that the final fragment to be ligated can contain the desired mutation.

Career Avenues

# GENE TRANSFER TECHNOLOGIES

## DNA/CALCIUM PHOSPHATE COPRECIPITATE METHOD

The ability of mammalian cells to take up exogenously supplied DNA from their culture medium was first reported by Szybalska and Szybalski (1962). They used total uncloned genomic DNA to transfect human cells deficient for the enzyme hypoxanthine guanine phosphoribosyl transferase (HPRT). Rare HPRT-positive cells, which had presumably taken up fragments of DNA containing the functional gene, were identified by selection on HAT medium. At this time, the actual mechanism of DNA uptake was not understood. Much later, it was appreciated that successful DNA transfer in such experiments was dependent on the formation of a fine DNA/calcium phosphate co-precipitate, which first settles on to the cells and is then internalized.

The precipitate must be formed freshly at the time of transfection. It is thought that small granules of calcium phosphate associated with DNA are taken up by endocytosis and transported to the nucleus, where some DNA escapes and can be expressed. The technique became applied to the analysis of the infectivity of adenoviral DNA. It is now established as a general method for the introduction of DNA into a wide range of cell types in culture. However, since the precipitate must coat the cells, this method is suitable only for cells growing in monolayers, not those growing in suspension or as clumps.

As originally described, calcium phosphate transfection was limited by the variable and rather low proportion of cells that took up DNA (1–2%). Only a small number of these would be stably transformed. Improvements to the method have increased the transfection frequency to 20% for some cell lines. A variant of the technique, using a different buffer system, allows the precipitate to form slowly over a number of hours, and this can increase stable transformation efficiency by up to 100-fold when using high-quality plasmid DNA.

## OTHER CHEMICAL TRANSFECTION METHODS

The calcium phosphate method is applicable to many cell types, but some cell lines are adversely affected by the co-precipitate due to its toxicity and are hence difficult to transfect. Alternative chemical transfection methods have been developed to address this problem. One such method utilizes diethylaminoethyl dextran (DEAE-dextran), a soluble polycationic carbohydrate that promotes interactions between DNA and the endocytotic machinery of the cell. This technique was initially devised to introduce viral DNA into cells but was later adapted as a method for plasmid DNA transfer. The efficiency of the original procedure was improved by adding after-treatments, such as osmotic shock or exposure to chloroquine, the latter having been shown to inhibit the acidification of endosomal vesicles. Although efficient for the transient transfection of many cell types, DEAE-dextran cannot be used to generate stably transformed cell lines. Another polycationic chemical, the detergent Polybrene, has been used for the transfection of Chinese hamster ovary (CHO) cells, which are not amenable to calcium phosphate transfection.

Career Avenues

## PHOSPHOLIPIDS AS GENE-DELIVERY VEHICLES

An alternative to chemical transfection procedures is to package DNA inside a phosopholipid vesicle, which interacts with the target cell membrane and facilitates DNA uptake. Initially bacterial protoplasts containing plasmids used to transfer DNA into mammalian cells. Briefly, bacterial cells were transformed with a suitable plasmid vector and then treated with chloramphenicol to amplify the plasmid copy number. Lysozyme was used to remove the cell walls, and the resulting protoplasts were gently centrifuged on to a monolayer of mammalian cells and induced to fuse with them, using polyethylene glycol. By using similar strategy haemoglobin-free ghosts of erythrocytes has been designed as delivery vehicles. The procedures are very efficient in terms of the number of transformants obtained, but they are also labour-intensive and so have not been widely adopted as a general transfection method. However, an important advantage is that they are gentle, allowing the transfer of large DNA fragments without shearing. Yeast cells with the cell wall removed (sphaeroplasts) have therefore been used to introduce yeast artificial chromosome (YAC) DNA into mouse ES cells by this method, for the production of YAC transgenic mice.

More widespread use has been made of artificial phospholipid vesicles, which are called *liposomes*. Initial liposome-based procedures were hampered by the difficulty encountered in encapsulating the DNA, and provided transfection efficiency no better than the calcium phosphate method. However, a breakthrough came with the discovery that cationic/neutral lipid mixtures can spontaneously form stable complexes with DNA that interact productively with the cell membrane, resulting in DNA uptake by endocytosis. This low-toxicity transfection method, commonly known as *lipofection*, is one of the simplest to perform and is applicable to many cell types that are difficult to transfect by other means, including cells growing in suspension. This technique is suitable for transient and stable transformation, and is sufficiently gentle to be used with YACs and other large DNA fragments. The efficiency is also much higher than that of chemical transfection methods – up to 90% of cells in a culture dish can be transfected. A large number of different lipid mixtures is available, varying in efficiency for different cell lines. A unique benefit of liposome gene-delivery vehicles is their ability to transform cells in live animals following injection into target tissues or even the bloodstream. Transfection efficiency has been improved and targeting to specific cell types achieved by combining liposomes with viral proteins that promote cell fusion, nuclear targeting signals and various molecular conjugates that recognize specific cell surface molecules. The development of liposomes for gene therapy has been comprehensively reviewed.

## ELECTROPORATION

Electroporation involves the generation of transient, nanometre-sized pores in the cell membrane, by exposing cells to a brief pulse of electricity. DNA enters the cell through these pores and is transported to the nucleus. The most critical parameters are the intensity and duration of the electric pulse, and these must be determined empirically for different cell types. However, once optimal electroporation parameters have been established, the method is simple

Career Avenues

to carry out and highly reproducible. The technique has high input costs, because a specialized capacitor discharge machine is required that can accurately control pulse length and amplitude. Additionally, larger numbers of cells may be required than for other methods because, in many cases, the most efficient electroporation occurs when there is up to 50% cell death. In an alternative method, pores are created using a finely focused laser beam. Although very efficient (up to 0.5% stable transformation), this technique is applicable to only small number of cells and has not gained widespread use.

## DIRECT TRANSFER METHODS

A final group of methods considered in this section encompasses those in which the DNA is transferred directly into the cell nucleus. One such procedure is microinjection, a technique that is guaranteed to generate successful hits on target cells but that can only be applied to a few cells in any one experiment. This technique has been applied to cultured cells that are recalcitrant to other transfection methods, but its principal use is to introduce DNA and other molecules into large cells, such as oocytes, eggs and the cells of early embryos. Particle bombardment is another direct delivery method, initially developed for the transformation of plants. This involves coating small metal particles with DNA and then accelerating them into target tissues using a powerful force, such as a blast of high-pressure gas or an electric discharge through a water droplet. In animals, this technique is most often used to transfect multiple cells in tissue slices rather than cultured cells. It has also been used to transfer DNA into skin cells *in vivo.*

Career Avenues

# GENE THERAPY

## GENE TRANSFER TO HUMANS – GENE THERAPY

### THE SCOPE OF GENE THERAPY

Gene therapy is any procedure used to treat disease by modifying the genetic information in the cells of the patient. In essence, gene therapy is the antithesis of the disease modeling discussed above. Whereas disease modeling takes a healthy animal and uses gene-manipulation techniques to induce a specific disease, gene therapy takes a diseased animal (or human) and uses gene-manipulation techniques in an attempt to correct the disorder and return the individual to good health. Gene transfer can be carried out in cultured cells, which are then reintroduced into the patient, or DNA can be transferred to the patient *in vivo*, directly or using viral vectors.

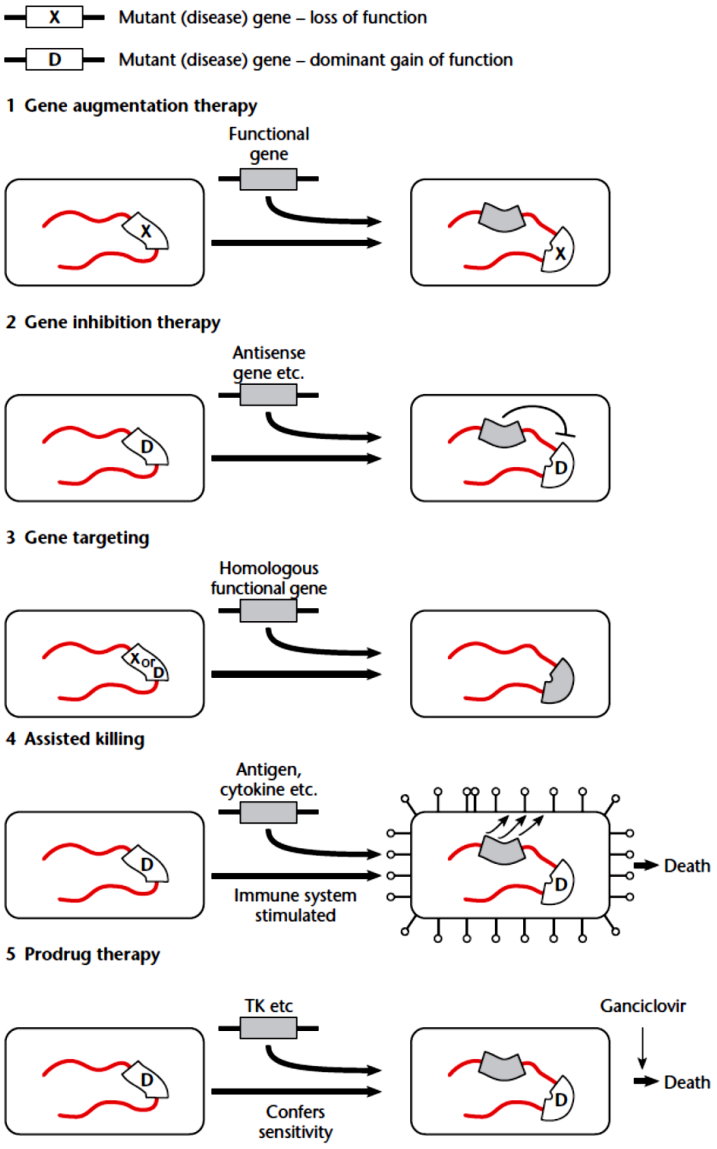


Fig. Overview of gene-therapy Strategies

The *ex vivo* approach can be applied only to certain tissues, such as bone marrow, in which the cells are amenable to culture. Gene therapy can be used to treat diseases caused by mutations in the patient's own DNA (inherited disorders, cancers), as well as infectious diseases, and is particularly valuable in cases where no conventional treatment exists or where that treatment is inherently risky. Strategies include the following:

- Gene-augmentation therapy (GAT), where DNA is added to the genome with the aim of replacing a missing gene product.
- Gene targeting to correct mutant alleles.
- Gene-inhibition therapy, using techniques such as antisense RNA expression or the expression of intracellular antibodies to treat dominantly acting diseases.
- The targeted ablation of specific cells.

Therapeutic gene transfer effectively generates transgenic human cell clones and, for this reason, only somatic cells can be used as targets. The prospect of germ-line transgenesis in humans raises serious ethical concerns and, with the rapid advances in technology allowing germ-line transformation and nuclear transfer in numerous mammals, these concerns will need to be addressed in the very near future. As an alternative to permanent gene transfer, transient gene therapy can be achieved using oligonucleotides, which can disrupt gene expression at many levels but do not permanently change the genetic material of the cell.

The tools and techniques for gene therapy are essentially similar to those used for gene transfer to any animal cells. Transfection, direct delivery or transduction can be used to introduce DNA into cells. Viral vectors are most popular because of their efficiency of gene transfer *in vivo*. However, extreme precautions need to be taken to ensure the safety of such vectors, avoiding potential problems, such as the production of infectious viruses by recombination and the pathological effects of viral replication. A number of viral vectors have been developed for gene therapy, including those based on oncoretroviruses, lentiviruses, adenovirus, adeno-associated virus, herpes virus and a number of hybrid vectors combining advantageous elements of different parental viruses. The risks associated with viral vectors have promoted research into other delivery methods, the most popular of which include direct injection of DNA into tissues (e.g. muscle), the injection of liposome–DNA complexes into the blood and direct transfer by particle bombardment. Although inherently much safer than viruses, such procedures show a generally low efficiency.

## GENE-AUGMENTATION THERAPY FOR RECESSIVE DISEASES

The first human genetic-engineering experiment was one of *gene marking*, rather than gene therapy, and was designed to demonstrate that an exogenous gene could be safely transferred into a patient and that this gene could subsequently be detected in cells removed from the patient.

Career Avenues

Both objectives were met. Tumour-infiltrating lymphocytes (cells that naturally seek out cancer cells and then kill them by secreting proteins such as tumour necrosis factor (TNF)) were isolated from patients with advanced cancer. The cells were then genetically marked with a neomycin-resistance gene and injected back into the same patient.

The first clinical trial using a therapeutic gene transfer procedure involved a 4-year-old female patient, Ashanthi De Silva, suffering from severe combined immune deficiency, resulting from the absence of the enzyme adenosine deaminase (ADA). This disease fitted many of the ideal criteria for gene-therapy experimentation. The disease was life-threatening (therefore making the possibility of unknown treatment-related side-effects ethically acceptable), but the corresponding gene had been cloned and the biochemical basis of the disease was understood. Importantly, since ADA functions in the salvage pathway of nucleotide biosynthesi, cells in which the genetic lesion had been corrected had a selective growth advantage over mutant cells, allowing them to be identified and isolated *in vitro*.

Conventional treatment for ADA deficiency involves bone-marrow transplantation from a matching donor. Essentially the same established procedure could be used for gene therapy, but the bone-marrow cells would be derived from the patient herself and would be genetically modified *ex vivo*. Cells from the patient were subjected to leucophaeresis and mononuclear cells were isolated. These were grown in culture under conditions that stimulated T-lymphocyte activation and growth and then transduced with a retroviral vector carrying a normal *ADA* gene as well as the neomycin-resistance gene. Following infusion of these modified cells, both this patient and a second, who began treatment in early 1991, showed an improvement in their clinical condition as well as in a battery of *in vitro* and *in vivo* immune-function studies. However, the production of recombinant ADA in these patients is transient, so each must undergo regular infusions of recombinant T lymphocytes. Research is ongoing into procedures for the transformation of bone-marrow stem cells, which would provide a permanent supply of corrected cells.

Gene-augmentation therapies for a small number of recessive single-gene diseases are now undergoing clinical trials. We consider cystic fibrosis (CF) as an example. CF is a disorder that predominantly affects the lungs, liver and pancreas. The disease is caused by the loss of a cAMP-regulated membrane-spanning chloride channel. This results in an electrolyte imbalance and the accumulation of mucus, often leading to respiratory failure. CF is a recessive disorder suggesting that the loss of function could be corrected by introducing a functional copy of the gene. Indeed, epithelial cells isolated from CF patients can be restored to normal by transfecting them with the cloned cystic-fibrosis transmembrane regulator (*CFTR*) cDNA. Unlike ADA deficiency, the cells principally affected by CF cannot be cultured and returned to the patient, so *in vivo* delivery strategies must be applied. Targeted delivery of the *CFTR* cDNA to affected cells has been achieved using adenoviral vectors, which have a natural tropism for the epithelial lining of the respiratory system. Recombinant viruses carrying the *CFTR* cDNA have been introduced into patients using an inhaler. The *CFTR* cDNA has also been introduced using liposomes. While such treatments have resulted in *CFTR* transgene expression in the nasal epithelium, there were

Career Avenues

neither consistent changes in chloride transport nor reduction in the severity of CF symptoms, i.e. they have been largely ineffective.

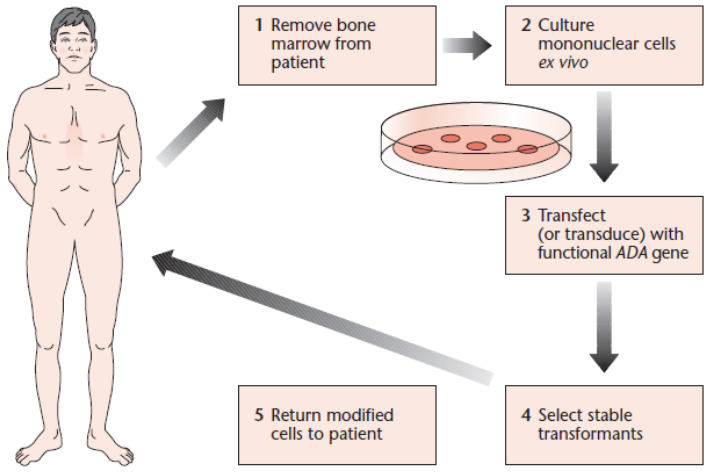



Fig. 14.10 Procedure for *ex vivo* gene therapy, based on the treatment for ADA deficiency.

## GATE BT 2010

Q.1 The bacteria known to be naturally competent for transformation of DNA is

(A) *Escherichia coli* (B) *Bacillus subtilis*
(C) *Mycobactrium tuberculosis* (D) *Yersinia pestis*

Q.2 Antibiotic resistance marker that CANNOT be used in a cloning vector in Gram negative bacteria is

(A) Streptomycin (B) Ampicillin
(C) Vancomycin (D) Kanamycin

Q.3 An example for template independent DNA polymerase is

(A) DNA Polymerase I (B) RNA polymerase
(C) Terminal deoxynucleotidyl transferase (D) DNA Polymerase III

Q.4 Expressed Sequence Tag is defined as

(A) a partial sequence of a codon randomly selected from cDNA library
(B) the characterstic gene expressed in the cell
(C) the protein coding DNA sequence of a gene
(D) uncharacterized fragment of DNA presence in the cell.

Career Avenues

Q.5 Consider the following statements.
I. T4 DNA ligase can catalyze blunt end ligation more efficiently than *E.coli* DNA ligase.
II. The ligation efficiency of T4 DNA ligase can be increased with PEG and ficoll.

(A) only I is true
(B) both I and II are true
(C) only II is true
(D) I is true and II is false

Q.6 Match the items in Group I with Group II.

| Group I | Group II |
|---|---|
| P. λ phage | 1. 35 – 45 kb |
| Q. Bacterial Artificial Chromosomes (BACs) | 2. 100 – 300 kb |
| R. Pl derived Artificial Chromosmes (PACs) | 3. <= 300kb |
| S. λ cosmid | 4. 5 – 25 kb |

(A) P-3,Q-4,R-1,S-2
(B) P-1,Q-3,R-2,S-4
(C) P-4,Q-3,R-2,S-1
(D) P-1,Q-2,R-3,S-4

Q.7 Match Group I with Group II.

| Group I | Group II |
|---|---|
| P. Real Time-PCR | 1. Biochips |
| Q. 2-D Electrophoresis | 2. Syber Green |
| R. Affinity chromatography | 3. Antibody linked sephrose beads |
| S. Microarray | 4. Ampholytes. |

(A) P-1,Q-2,R-4,S-3
(B) P-2,Q-3,R-4,S-1
(C) P-2,Q-4,R-3,S-1
(D) P-3,Q-2,R-1,S-4

## GATE BT 2011

Q.8 Yeast artificial chromosome (YAC's) are used for cloning

(A) large segments of DNA
(B) mRNA
(C) bacterial DNA
(D) yeast DNA

Q.9 Restriction endonucleases which recognize and cut same recognition sequences are known as

(A) isoschizomers
(B) isozymes
(C) isoaccepting endonucleases
(D) abzymes

Career Avenues

Q.10 Protein-protein interactions are studied by

P. DNA foot printing
Q. Yeast two hybrid system
R. Ligase chain reaction
S. Mass spectrometry

(A) P and S only
(B) Q and S only
(C) P and R only
(D) Q and R only

Q.11 Identify the **CORRECT** statements

P. 5' and 3'ends of the transcripts can be mapped by utilizing polymerase chain reaction.
Q. $S_1$ nuclease can cleave the DNA strand of a DNA-RNA hybrid.
R. $T_4$ polynucleotide kinase is used for labeling 3' end of DNA.
S. Baculovirus (*Autographa californica*) can be used as an insect expression vector.

(A) P and Q only
(B) R and S only
(C) P and S only
(D) Q and R only

A 200 µl of polymerase chain reaction has 100 template DNA molecules and the reaction performed for 10 cycles.

Q.12 How many molecules of amplicons will be generated?

(A) $1.024 \times 10^4$ (B) $1.024 \times 10^5$ (C) $2.048 \times 10^4$ (D) $2.048 \times 10^5$

Q.13 How many molecules of amplicons will be present in 0.1 µl of reaction ?

(A) 102.4 (B) 1024 (C) 51.2 (D) 512

## GATE BT 2012

Q.14 The basis for blue-white screening with pUC vectors is

(A) intraallelic complementation
(B) intergenic complementation
(C) intragenic suppression
(D) extragenic suppression

Q.15 Protein-DNA interactions *in vivo* can be studied by

(A) gel shift assay
(B) Southern hybridization
(C) chromatin immunoprecipitation assay
(D) fluorescence *in situ* hybridization assay

Career Avenues

Q.16 Synteny refers to

(A) gene duplication from a common ancestor
(B) a tree representation of related sequences
(C) the extent of similarity between two sequences
(D) local conservation of gene order

**Answers:**

- **Q.1 – B.**
- **Q.2 – C.**
- **Q.3 – C.**
- **Q.4 – C.**
- **Q.5 – B.**
- **Q.6 – C.**
- **Q.7 – C.**
- **Q.8 – A.**
- **Q.9 – A.**
- **Q.10 – B.**
- **Q.11 – C.**
- **Q.12 – B.**
- **Q.13 – C.**
- **Q.14 – A.**
- **Q.15 – D.**
- **Q.16 – A.**

Career Avenues